

THE DOLMEN OF MENGA

A Complete Independent Investigation into the Oldest Great Megalith of Iberia

The stone, the mountain, the light, and the water — an evidence-led examination of a 5,700-year-old engineering project on the plain of Antequera, Andalusia, Spain.

FOREWORD: HOW THIS INVESTIGATION WORKS

This book began as an investigation file and it keeps that discipline throughout. The subject is the Dolmen of Menga — a vast stone chamber built near the town of Antequera, in the province of Málaga, southern Spain, between roughly 3800 and 3600 BCE — and everything that surrounds it: the hill it sits on, the quarry its stones came from, the mountain its corridor points at, the well sunk into its floor, and the nearly six millennia of human behavior that followed its construction.

Three ground rules govern every page.

First rule: evidence carries a pedigree. Every factual claim in this book is tagged so the reader always knows what kind of knowledge it is:

- **[MEASURED]** — obtained by instrument: laser scanning, geological mapping, petrography, survey, radiocarbon or luminescence dating. The strongest tier.
- **[DOCUMENTED]** — recorded in excavation reports, historical documents, or peer-reviewed literature.
- **[INFERRED]** — a conclusion logically drawn from measured or documented evidence by the researchers who produced it, or by this investigation, and explicitly labeled as reasoning rather than observation.
- **[REPORTED]** — attested in historical or traditional sources (19th-century accounts, legends, early descriptions) that cannot be independently re-verified today.
- **[OPEN]** — genuinely unresolved. The honest frontier.

Second rule: no adopted hypotheses. This investigation reads the hypotheses of others — archaeologists, geologists, archaeoastronomers, folklorists, and independent researchers — but adopts none of them as its own. Each hypothesis is presented with its evidence and its weaknesses. Where the data force a conclusion, the conclusion is stated as this investigation's own finding, with the chain of reasoning shown. Where the data do not force a conclusion, the question is left open and marked [OPEN].

Third rule: follow the clues, not the parties. This investigation is not aligned with any school — neither the “Neolithic genius” celebration nor the “primitive farmers” minimization, neither orthodoxy nor fringe. The anomalies are listed because they are measured facts that deviate from the norm. What they mean is a separate question, and it is treated as such.

A glossary at the end of the book explains every technical term in plain language. No prior knowledge of archaeology is assumed. Wherever a number is given — a weight, an angle, a depth, a date — it is given because someone measured it, and the source of the measurement is named.

PROLOGUE: THE HILL OF STONES

Approach Antequera from the south, across the flat green chessboard of the Guadalhorce valley, and two things claim the eye before the town does.

The first is a mountain that should not look the way it does. La Peña de los Enamorados rises 880 meters above sea level out of an otherwise gentle plain, a limestone massif elongated north to south, and from the west — particularly in low morning or evening light — it does not look like a mountain at all. It looks like a face. A human profile, lying on its back, chin pointed at the sky: forehead, nose, lips, and a jutting chin, drawn in 100-meter cliffs of oolitic and nodular limestone [MEASURED]¹. People have called it the Sleeping Giant, the Indian's Head, the Lovers' Rock. Every culture that has lived on this plain has had a name for it and a story about it.

The second is a hill that is not a hill. About a kilometer and a half from the mountain's northern face, on the northeastern edge of modern Antequera, stands a broad, grass-covered mound roughly 48 by 44 meters across [MEASURED]². It looks like a natural rise in the terrain until you walk through the passage cut into its northeastern side and find yourself inside a tunnel of stone.

The tunnel is about 25 meters long [MEASURED]³. Its walls are single slabs of rock — ten on each side, each one several meters tall and weighing tens of tons. They do not stand vertically. They lean inward, deliberately, at an average of about five degrees off the vertical, so the chamber is narrower at the ceiling than at the floor. The ceiling is five more slabs, the largest of them weighing about 150 tons — roughly the weight of a fully grown blue whale, or of thirty mid-size cars [MEASURED]⁴. Down the center of the chamber, three colossal pillars hold the joints where the roof slabs meet. At the far end, a single backstone closes the space. And in the floor of the rear third of the chamber, aligned precisely with the central axis, a circular shaft drops 19.4 meters straight down through the bedrock into the groundwater below [MEASURED]⁵.

The whole thing was then buried under thousands of tons of carefully layered earth and stone — the “hill” you walked across to get in.

This is the Dolmen of Menga. It was built between about 3800 and 3600 BCE [MEASURED]⁶, which makes it older than the Great Pyramid of Giza by more than a thousand years, older than Stonehenge's sarsen circle by well over a millennium, older than writing, older than the wheel in this part of the world, older than metal tools. The people who built it had no copper, no bronze, no iron, no draft animals, no pulleys, no mathematics written down. They had stone, wood, rope, water, fire, and time — and they

¹García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña's geology and the 45° alignment's cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

²García Sanjuán, L., et al. — author manuscript of the multimethod dating paper (University of Southampton ePrints), including mound dimensions and the catalog of historical radiocarbon determinations. https://eprints.soton.ac.uk/470780/1/Garcia_Sanjuan_et_al_2022_A_multimethod_approach_to_the_genesis_of_menga_a_world_heritage_megalith.pdf

³Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

⁴Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁵García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

⁶García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

produced what the geologists who studied it most recently called “a unique example of creative genius and early science among Neolithic societies... designed as a completely original engineering project, for which we know of no precedents in Iberia” [DOCUMENTED]⁷.

That phrase — *no known precedents* — is where this investigation begins. Because Menga is not merely impressive. It is, in a series of specific, measurable ways, anomalous. It breaks the statistical rules of its own culture. It faces the wrong direction. It is built of the wrong stone. Its stones are buried to depths that make no structural sense until you calculate earthquakes into the equation. It contains a hydraulic feature with no known parallel in the megalithic world. And it sits at one end of a straight line drawn across the landscape that took a thousand years to complete.

This book examines every one of those anomalies in full detail — what is measured, what is inferred, what is claimed, what is contested, and what remains genuinely unexplained. But before we can read the monument, we have to read the ground it stands on. Every anomaly at Menga is a dialogue between a structure and its landscape, and the landscape came first.

PART I — THE GROUND BEFORE THE STONES

CHAPTER 1: THE PLAIN AND ITS PROBLEM WITH WATER

1.1 The crossroads of Andalusia

The Antequera plain — the Vega de Antequera — is a broad, fertile depression in the geographic heart of Andalusia, sitting at roughly 400 to 450 meters above sea level, ringed by mountains: the El Torcal karst massif to the south (rising to 1,100–1,400 m), the Sierra de las Nieves and the Subbetic ranges beyond, and the limestone ridge of La Peña de los Enamorados immediately to the northeast [MEASURED]⁸.

Position is the first fact about this place. Antequera sits at a natural crossroads where four corridors of travel meet: south to Málaga and the Mediterranean coast; north through the mountain passes toward Córdoba and the interior of the Iberian Peninsula; west down the Guadalquivir valley toward Seville and the Atlantic; east toward Granada and the Spanish Levant [DOCUMENTED]⁹. In an age when all travel was on foot, a valley where four routes cross is a place where people, animals, objects, and ideas accumulate. This is not speculation — it is geography, and it is corroborated by the archaeological record: the Antequera plain holds one of the densest concentrations of Neolithic and Copper Age settlement evidence in southern Iberia, including the sites of Arroyo Saladillo, Huerta del Ciprés, El Silillo, El Perezón, and Piedras Blancas, several of them excavated under modern conditions and radiocarbon dated [DOCUMENTED]¹⁰.

⁷University of Alcalá repository record for the *Science Advances* paper, including the official abstract (“no precedents in Iberia”). <https://ebuah.uah.es/dspace/handle/10017/63692?show=full>

⁸García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

⁹García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

¹⁰Museos de Andalucía — official site brochure (English): research history from 1530 onward, El Toro sequence, settlement catalog, Hoskin’s 2002 confirmations. <https://www.museosdeandalucia.es/documents/1971642/0/Folieto+en+ingl%C3%A9s/>

1.2 The land of brackish water

The second fact about this place is less obvious on a map, and it will matter enormously when we reach the deepest anomaly inside the dolmen: **the Antequera region has a water problem.**

Beneath the plain, the local geology is dominated by Triassic formations rich in gypsum, halite (common rock salt), and sylvite (potassium chloride) [DOCUMENTED]¹¹. Where groundwater moves through these formations and emerges as springs, it arrives loaded with calcium sulfate and sodium chloride — chemically “poor” water, brackish, unpalatable, and marginal for human consumption. The regional toponymy records this in plain language: the map of the Antequera district is dotted with place names referring to salt and brine, and twenty kilometers away lies the Laguna de Fuente de Piedra, a salt lake famous today for its flamingos, the largest natural lake in Andalusia [DOCUMENTED]¹².

Fresh water on this plain is not the default condition. It is a localized, valuable exception. The main reliable fresh sources are the La Villa river, fed by the El Nacimiento spring at the edge of the town, and — critically — the aquifer beneath a particular hill on the plain’s northeastern side [DOCUMENTED]¹³.

A community of farmers deciding where to settle, where to gather, and where to build something meant to last forever would know, to the meter, where fresh water could be reached. Water was not a convenience; it was the infrastructure of survival and, very probably, of ritual. Keep this fact in storage. It returns with full force in Chapter 13.

1.3 The two natural monuments

Two geological formations dominate the mental landscape of the plain, and both were later drawn into the architectural project of the dolmens.

El Torcal, eleven kilometers south of the town, is a karst plateau of Jurassic limestone, laid down on the bed of the Tethys Ocean 150 to 200 million years ago and dissolved by water into a forest of surreal stacked pillars and corridors [DOCUMENTED]¹⁴. To a pre-scientific mind, El Torcal does not look like geology. It looks like ruins — like a city built by giants and abandoned. Its caves were occupied from the very beginning of the Neolithic, and its slopes provided flint, marble, ophite, dolerite, iron oxides, and salt — a natural warehouse of tool-stone and pigment [DOCUMENTED]¹⁵.

La Peña de los Enamorados, already described in the Prologue, is the second: an 880-meter limestone massif whose western profile reads unmistakably as a recumbent human face [MEASURED]¹⁶. Its northern face terminates in a 100-meter-high vertical cliff of protruding oolitic limestone that forms the “chin” of the sleeping figure — and this specific cliff plane, as we will see, is the exact target of Menga’s axis

¹¹García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well’s excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

¹²García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well’s excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

¹³García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well’s excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

¹⁴Go-Torcal-Antequera — UNESCO site guide: the four-event astronomical cycle; Viera equinox; El Romeral 199°/winter solstice. <https://www.go-torcal-antequera.com/en/the-el-torcal/unesco-world-heritage-antequera-dolmens-site/>

¹⁵García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

¹⁶García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

[MEASURED]¹⁷.

The UNESCO inscription of 2016 formally recognized what the prehistoric builders had encoded: the Antequera Dolmens Site is the only World Heritage property in which two *natural* monuments (La Peña and El Torcal) are inscribed alongside three built ones, precisely because the built monuments were designed in dialogue with the natural ones [DOCUMENTED]¹⁸. UNESCO’s own wording calls Menga “the only dolmen in continental Europe that faces towards an anthropomorphic mountain” [DOCUMENTED]¹⁹.

1.4 The hill itself

The hill the dolmen was built into is a low rise of calcarenite — a sedimentary rock of cemented sand-sized carbonate grains — on the northeastern edge of the modern town, beside the old road to Archidona. It is not the highest ground on the plain, nor the most defensible. Its qualifications were different, and they emerge piece by piece through this book: it stood about a kilometer from the best available stone [MEASURED]²⁰; it stood over an accessible aquifer [MEASURED]²¹; and it occupied — and this was established by geometric analysis — the one position from which a chamber axis could simultaneously lock onto the northern cliff face of La Peña and produce a specific, reproducible light effect inside the chamber at the summer solstice [INFERRED from MEASURED geometry]^{22,23}.

Before the first stone was cut, the hill was already busy. The basal layer of the mound — a 1.5-meter-thick stratum of very dark soil sealed beneath the tumulus — yielded large quantities of hand-made pottery and knapped flint alongside abundant organic material, all radiocarbon dated to the Late Neolithic, recording intense human activity on the hill before and during construction [MEASURED]²⁴. The builders did not choose an empty stage. They chose a place that was already a place.

CHAPTER 2: THE FIRST FARMERS — WHO WAS HERE, AND WHERE DID THEY GO?

2.1 The cave that housed a world

The human story of the Antequera plain in the Neolithic begins, as so many Iberian stories do, in a cave.

¹⁷García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

¹⁸UNESCO World Heritage Centre — “Antequera Dolmens Site,” inscription 1501 (2016), criteria (iii) and (iv). <https://whc.unesco.org/en/list/1501/>

¹⁹UNESCO World Heritage Centre — “Antequera Dolmens Site,” inscription 1501 (2016), criteria (iii) and (iv). <https://whc.unesco.org/en/list/1501/>

²⁰Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

²¹García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²²NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

²³García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

²⁴García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

El Toro cave, high in the El Torcal massif, was occupied from the very start of the Neolithic in this region — around 5400 BCE — by some of the first farming communities to spread along the Mediterranean coast of Iberia [MEASURED, by a series of 29 radiocarbon determinations]²⁵. For more than a thousand years, the cave served as dwelling, animal pen, and very likely ritual space. Its occupants herded sheep and goats, grew wheat and barley, knapped flint from the Torcal's own seams, and made pottery by hand. After an interruption of several centuries, occupation intensified again between roughly 4300 and 3800 BCE, a period of strong growth in farming, herding, and craft activity across the whole district [DOCUMENTED]²⁶.

These cave communities, in the assessment of the researchers who have studied them longest, are the direct cultural predecessors of the people who would build the megaliths [DOCUMENTED]²⁷.

Then the cave died.

2.2 The collapse

Bayesian modeling of the radiocarbon series from El Toro places the end of its occupation with remarkable precision: the cave was abandoned between roughly 4250 and 3935 BCE (2 range), most probably around 4000 BCE [MEASURED]²⁸. And the cause appears to have been violent. A massive roof slab collapsed inside the cave — an event dated by the same modeling exercise — rendering the space unusable [MEASURED/INFERRED]²⁹.

Consider what this meant. A place that had sheltered human life for fourteen centuries — the anchor of a community's world, holding its dead, its stories, its sense of continuity — was destroyed in a single geological instant. The research team behind the dating study frames it carefully: “the sudden abandonment of the cave, most likely caused by the catastrophic collapse of the roof slab, must have resonated across the region” [INFERRED]³⁰.

This investigation flags the timing without over-reading it: **within a few centuries of the roof of their world falling in, the people of this plain began building a stone roof of their own — one engineered, as we will see, with an almost obsessive redundancy against collapse.** The chamber of Menga has five capstones, four internal support pillars (originally), walls interlocked like puzzle pieces, foundations sunk three meters into bedrock, and a relieving arch distributing the roof load outward. Whether the El Toro trauma and the Menga engineering are causally linked is [OPEN]. That they are chronologically adjacent is [MEASURED].

²⁵García Sanjuán, L., et al. — author manuscript of the multimethod dating paper (University of Southampton ePrints), including mound dimensions and the catalog of historical radiocarbon determinations. https://eprints.soton.ac.uk/470780/1/Garcia_Sanjuan_et_al_2022_A_multimethod_approach_to_the_genesis_of_menga_a_world_heritage_megalith.pdf

²⁶Museos de Andalucía — official site brochure (English): research history from 1530 onward, El Toro sequence, settlement catalog, Hoskin's 2002 confirmations. <https://www.museosdeandalucia.es/documents/1971642/0/Folieto+en+ingl%C3%A9s/>

²⁷Museos de Andalucía — official site brochure (English): research history from 1530 onward, El Toro sequence, settlement catalog, Hoskin's 2002 confirmations. <https://www.museosdeandalucia.es/documents/1971642/0/Folieto+en+ingl%C3%A9s/>

²⁸García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²⁹García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

³⁰García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

2.3 The world outside the cave

By the early fourth millennium BCE, life on the plain had moved decisively into the open. Excavations at Arroyo Saladillo (6 km west of Menga) and Huerta del Ciprés (barely 700 meters north of the dolmen) document farming settlements with large quantities of knapped lithics and grinding tools — the signature of intensifying agriculture and a growing human footprint on the land [DOCUMENTED]³¹.

Two findings from this horizon matter for what follows.

At **Arroyo Saladillo**, an individual was buried between 4037 and 3805 BCE (2) beneath a cairn of medium and large stones, probably combined with a ditch — one of the earliest proto-megalithic graves in Iberia [MEASURED]³². This is the local “before” picture: the scale of funerary architecture on the plain in the generations immediately preceding Menga. A cairn of stones over a single grave.

At **La Peña de los Enamorados**, on the mountain’s northern face at the foot of the great cliff, the Matacabras rock shelter was already in use before about 3800 BCE, its walls painted with schematic-style motifs [MEASURED, by radiocarbon and multi-analytical study]³³³⁴. A sanctuary, in effect, already marked the mountain before the dolmen existed.

Hold these two facts side by side, because together they define the central anomaly of scale that this investigation will keep returning to. Within living memory of a single-grave cairn, the same society produced a 1,140-ton interlocked stone chamber [MEASURED]³⁵. And it aimed that chamber — with a precision we will measure — at a mountain sanctuary that already existed.

2.4 What kind of society?

No written record names these people. No skeleton wears a crown. What we know of them is structural: their settlement density, their tools, their exchange networks, and above all the monument itself, read as a fossil of organization.

What Menga requires of a society is calculable [INFERRED from MEASURED data]:

- **Labor:** 32 stones averaging over 35 tons each, the largest near 150 tons, quarried, moved a kilometer, and set with millimetric fits. Experimental work on megalith transport suggests moving loads of this class on wooden sledges over prepared trackway demands coordinated pulling teams numbering in the hundreds for the largest pieces, plus quarry crews, trackway crews, rope-makers, timber-cutters, and the food logistics to feed them.
- **Knowledge:** petrographic discrimination (choosing workable stone beds), fracture mechanics (exploiting tectonic joints), geodesy (the 45° axis), geometry (constant lean angles), hydrology (the well), and load engineering (the arch) — transmitted across generations without writing.

³¹García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

³²García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

³³García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

³⁴Wikipedia — “Dolmen of Menga” (reference summary; the “several hundred skeletons” tradition is [REPORTED] through this and the following). https://en.wikipedia.org/wiki/Dolmen_of_Menga

³⁵Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

- **Duration:** the Bayesian construction window is 140–275 years at 1 [MEASURED]³⁶ — meaning the project was likely not one generation’s work but a multi-generational program, or at minimum a program planned across generations.

A society that can do this is not a band of scattered homesteaders. It is a network of communities capable of pooling labor across the plain — the interpretation the site’s own management literature favors: “close cooperation between communities that shared common religious codes” [DOCUMENTED]³⁷. This investigation adopts the observation (pooled labor across communities) without adopting any specific model of hierarchy, chiefdom, or priesthood — the stones require coordination; they do not, by themselves, require kings [OPEN].

CHAPTER 3: THE MOUNTAIN WITH A FACE

3.1 Reading La Peña

La Peña de los Enamorados deserves a full chapter because, as the evidence will show, it is not scenery. It is a component of the monument.

The massif is limestone, oriented north–south, peaking at about 878–880 meters above sea level [MEASURED]^{38,39}. From the west and southwest — which is to say, from the plain, from the dolmen’s hill, from every settlement we know of — its skyline describes a recumbent human profile facing upward: forehead at the southern end, nose, lips, and a pronounced chin at the northern end [MEASURED/DOCUMENTED]⁴⁰. The effect is strongest at low sun, when shadow modeling sharpens the profile. The traditional Spanish name, *Peña de los Enamorados* (“Lovers’ Rock”), is medieval; the local alternative names — *Montaña del Indio*, the Sleeping Giant — record the same perception: a person, lying down, permanent, gigantic [DOCUMENTED]⁴¹.

The “chin” of the figure is geologically specific: a cliff roughly 100 meters high on the northern face, formed where oolitic limestone meets more erodible red nodular limestone, the harder beds protruding as a plane [MEASURED]⁴². This cliff — not the summit, not the profile generally, but this exact plane — is what Menga’s central axis, extended northeast across the plain, strikes [MEASURED]⁴³.

³⁶García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

³⁷Museos de Andalucía — official site brochure (English): research history from 1530 onward, El Toro sequence, settlement catalog, Hoskin’s 2002 confirmations. <https://www.museosdeandalucia.es/documents/1971642/0/Folieto+en+ingl%C3%A9s/>

³⁸Antequera World Heritage — La Peña de los Enamorados: morphology and the Tello–Tagzona legend. <https://www.antequerapatrimoniomundial.com/en/lovers-rock/>

³⁹García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

⁴⁰García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

⁴¹Kiddle — “Peña de los Enamorados”: legend variants, Southey’s *Laila and Manuel*, Columbus’s 1492 comparison. https://kids.kiddle.co/Pe%C3%B1a_de_los_Enamorados

⁴²García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

⁴³García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the->

At the foot of the chin, in a shallow shelter, are the Matacabras paintings.

3.2 Matacabras: the sanctuary that was already there

The Abrigo de Matacabras is a small rock shelter at the base of La Peña's northwestern face, containing schematic-style rock art — painted motifs in the abstract vocabulary of Iberian post-Paleolithic art. A multidisciplinary study published in 2018 (ATLAS group, University of Seville) established, by radiocarbon dating and pigment analysis, that the shelter was in use before about 3800 BCE — that is, *before Menga was built* [MEASURED]⁴⁴⁴⁵. The study concluded the shelter was “a place of reference for the Neolithic (and possibly even older) population of the region” [DOCUMENTED]⁴⁶.

The sequence matters: **the mountain was marked as significant before the monument was aimed at it**. Menga did not create the mountain's sanctity; it enlisted it [INFERRED from MEASURED chronology].

3.3 Piedras Blancas: the tomb in the giant's chest

In 2020, excavations at the site of Piedras Blancas — on a low calcarenite hill immediately below Matababras, in the fold of terrain that reads, from a distance, as the sleeping giant's *chest* — uncovered a previously unknown megalithic tomb, published in *Antiquity* in 2023 [DOCUMENTED]⁴⁷⁴⁸.

What they found defies easy categorization. The excavators describe it as “part natural, part built, part hypogeum, part megalith” [DOCUMENTED]⁴⁹: a pseudo-rectangular cavity 4.5 meters long and 1.45 meters wide, cut directly into bedrock, its long sides formed not by placed stones but by the natural vertical strata of an anticline fold, exploited deliberately [MEASURED]⁵⁰. Slabs at the entrance were dressed and engraved. One back slab appears to have been chosen precisely because its natural ripple marks — frozen seabed waves from the Eocene — resemble scales or water under raking light [MEASURED/INFERRED]⁵¹. An identical ripple-marked stone was used as a jamb at the threshold of the main chamber of the El Romeral tholos, three kilometers away — evidence of a shared design vocabulary across the complex [MEASURED]⁵².

antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB

⁴⁴Wikipedia — “Dolmen of Menga” (reference summary; the “several hundred skeletons” tradition is [REPORTED] through this and the following). https://en.wikipedia.org/wiki/Dolmen_of_Menga

⁴⁵García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

⁴⁶Wikipedia — “Dolmen of Menga” (reference summary; the “several hundred skeletons” tradition is [REPORTED] through this and the following). https://en.wikipedia.org/wiki/Dolmen_of_Menga

⁴⁷García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña's geology and the 45° alignment's cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

⁴⁸ArkeoNews — Piedras Blancas reportage. <https://arkeonews.net/archaeologists-discover-a-new-megalithic-monument-in-the-heart-of-andalusia-in-southern-spain-a-5000-year-old-secret/>

⁴⁹García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña's geology and the 45° alignment's cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

⁵⁰García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña's geology and the 45° alignment's cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

⁵¹García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña's geology and the 45° alignment's cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

⁵²García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña's geology and the 45° alignment's cliff-plane target.

Most striking: two triangular dressed stones were fixed to the chamber floor with mud mortar, oriented to the summer solstice sunrise, arranged to *funnel* the dawn light toward the back of the chamber [MEASURED]⁵³. The tomb was used in three major phases across more than a millennium before being abandoned between roughly 1950 and 1180 BCE [MEASURED]⁵⁴.

The excavators' conclusion is worth quoting because it names the pattern this whole chapter documents: the architecture of the Antequera plain works by “coupling the carving of rocks... with the natural orientation of the geological substrate to ‘domesticate’ sunlight” [DOCUMENTED]⁵⁵. At Piedras Blancas they domesticated it with two small triangles. At Menga, with 32 megaliths. The idea is the same; only the scale differs by three orders of magnitude.

3.4 The geographical coincidence that is not a coincidence

La Peña occupies a second, purely practical significance: it is the most visible landmark at the Antequera crossroads, a natural navigation beacon on the route between the coast and the interior [DOCUMENTED]⁵⁶. A sacred mountain that is also a wayfinding mountain concentrates traffic, and traffic concentrates meaning.

This investigation therefore records, as a working frame (not an adopted hypothesis), the regional research consensus that La Peña functioned as a “geo-sculpture” — a natural formation read as an artifact, around which activity, art, burial, and eventually monumentality accreted [DOCUMENTED]⁵⁷. Whether the people of 3800 BCE saw an ancestor, a god, a corpse, or a sleeping giant in that profile, we cannot know [OPEN]. What we can measure is that they built toward it, repeatedly, across centuries, with the largest stones available to them.

CHAPTER 4: THE IDEA OF THE MEGALITH — AND HOW MENGA BREAKS ITS RULES

4.1 A short history of big stones

Megalithic architecture — monuments built of large stones — is the earliest form of monumental stone construction in Europe [DOCUMENTED]⁵⁸. The phenomenon began in the fifth millennium BCE on the Atlantic façade of France: by about 4700 BCE, communities in Brittany were raising gigantic earthen mounds (the Carnac-region tumuli of Saint-Michel, Le Moustoir, Tumiach, 100–200 m long) over cen-

<https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

⁵³HeritageDaily — Piedras Blancas reportage: ripple-marked slabs, solstice-funnel stones. <https://www.heritagedaily.com/2023/05/complex-funerary-monument-found-at-foot-of-la-pena-de-los-enamorados/147231>

⁵⁴Miami Herald — Piedras Blancas reportage: use phases and abandonment 1950–1180 BCE. <https://www.miamiherald.com/news/nation-world/world/article275052621.html>

⁵⁵García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

⁵⁶García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

⁵⁷García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

⁵⁸University of Alcalá repository record for the *Science Advances* paper, including the official abstract (“no precedents in Iberia”). <https://ebuah.uah.es/dspace/handle/10017/63692?show=full>

tral stone-built tombs, and erecting menhirs — single standing stones, some of enormous size [DOCUMENTED]⁵⁹.

The single largest known megalithic stone in Europe is Breton: the Grand Menhir Brisé at Locmariaquer, a broken standing stone estimated at about 335 tons [MEASURED]⁶⁰. Keep that number; it is the benchmark against which Menga's capstone is second.

The megalithic idea reached Iberia later. The first proto-megalithic burial chambers in the Peninsula appear around 4200–4000 BCE [MEASURED]⁶¹ — the Arroyo Saladillo cairn belongs to exactly this horizon. Then, in the first half of the fourth millennium, the phenomenon exploded across western Iberia: thousands of passage graves (*antas* in Portugal, *dólmenes* in Spain), stone chambers with entrance corridors, covered by mounds.

4.2 Hoskin's law

In the late twentieth century, the Cambridge astrophysicist-turned-archaeoastronomer Michael Hoskin did something nobody had done: he went out with a theodolite and measured the orientations of megalithic tombs by the thousand — ultimately over 3,000 monuments across the western Mediterranean and Atlantic Europe [DOCUMENTED]⁶².

His Iberian result is one of the most robust statistical patterns in all of archaeoastronomy. Of 334 tombs Hoskin measured in western Iberia, 324 — 97.0% — faced within the azimuth range 60°–130°, the arc of sunrise (or just south of it, explainable by local horizon altitude) [MEASURED]⁶³. Widening the net to 945 tombs across Spain, Portugal, and the Pyrenean fringe, 911 — 96.4% — faced the sun rising or climbing the sky (range 60°–190°) [MEASURED]⁶⁴. Hoskin concluded that the custom was celestial and probably tied the tomb's facing to sunrise on the day construction began [INFERRED by Hoskin]⁶⁵.

Call it Hoskin's law: **Iberian passage graves face the rising sun**. It holds to a level of statistical regularity (96–97%) that is vanishingly rare in human affairs. It is the single most reliable predictive rule in Iberian megalithism.

Menga faces 45° [MEASURED]⁶⁶.

⁵⁹García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

⁶⁰Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁶¹García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

⁶²Hoskin, M. — “Orientations of dolmens of Western Europe,” *Complutum* 20(2), 165–175 (2009). The statistical law of Iberian tomb orientations (97% SR; 96.4% SR/SC). <https://revistas.ucm.es/index.php/CMPL/article/download/CMPL0909220165A/29024/30185>

⁶³Hoskin, M. — “Orientations of dolmens of Western Europe,” *Complutum* 20(2), 165–175 (2009). The statistical law of Iberian tomb orientations (97% SR; 96.4% SR/SC). <https://revistas.ucm.es/index.php/CMPL/article/download/CMPL0909220165A/29024/30185>

⁶⁴Hoskin, M. — “Orientations of dolmens of Western Europe,” *Complutum* 20(2), 165–175 (2009). The statistical law of Iberian tomb orientations (97% SR; 96.4% SR/SC). <https://revistas.ucm.es/index.php/CMPL/article/download/CMPL0909220165A/29024/30185>

⁶⁵Hoskin, M. — “Orientations of dolmens of Western Europe,” *Complutum* 20(2), 165–175 (2009). The statistical law of Iberian tomb orientations (97% SR; 96.4% SR/SC). <https://revistas.ucm.es/index.php/CMPL/article/download/CMPL0909220165A/29024/30185>

⁶⁶García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

Forty-five degrees is not inside the sunrise arc. It is about ten degrees *north of midsummer sunrise* — outside the range in which the sun rises at any time of year [MEASURED]⁶⁷. In a dataset where 97 of every 100 tombs obey the solar rule, the largest, earliest, and most engineered tomb of them all points where no sun ever rises.

Hoskin himself, who visited and measured the Antequera monuments, confirmed in 2002 that Menga’s orientation is terrestrial, not celestial: it targets the mountain [DOCUMENTED]⁶⁸. UNESCO’s inscription text repeats the finding: Menga is “the only dolmen in continental Europe that faces towards an anthropomorphic mountain” [DOCUMENTED]⁶⁹.

The anomaly is not that an explanation exists. The anomaly is the exception itself: the culture that built Menga knew the rule — its neighbors, predecessors, and successors obeyed it for two thousand years — and at the moment of maximum ambition, they broke it. Chapter 11 investigates what they got in exchange.

4.3 The three traditions at Antequera

The Antequera complex eventually comprised three major monuments, and each embodies a different architectural lineage [DOCUMENTED]⁷⁰:

- **Menga** (c. 3800–3600 BCE): a “simple gallery” dolmen — one vast chamber, no distinct corridor, walls of colossal orthostats, lintelled roof, interior pillars. The oldest and largest.
- **Viera** (immediately beside Menga, built later): a classic passage grave — long narrow corridor into a small rectangular chamber, oriented roughly east, into the equinoctial sunrise arc [MEASURED]⁷¹. The rule-abiding sibling.
- **El Romeral** (about 2–4 km from the others, built around a millennium after Menga, in the Copper Age): a *tholos* — a round chamber with a corbelled false-dome of inward-stepping masonry rings, Mediterranean in tradition, its long corridor aimed southwest at the Camorro de las Siete Mesas peak in El Torcal [DOCUMENTED]^{72,73}.

One plain, three monuments, three construction grammars, three orientations, a thousand years of activity. And — as Chapter 18 will lay out — they stand on a straight line [DOCUMENTED]⁷⁴.

End of Part I. Part II opens the quarry.

⁶⁷García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

⁶⁸Antequera Tourism — official monument dossier: the two-line orientation system (Menga→La Peña; Viera→equinox; El Romeral on the axis, → El Torcal). <https://turismo.antequera.es/wp-content/uploads/2011/06/DOLMENES-INGLES.pdf>

⁶⁹UNESCO World Heritage Centre — “Antequera Dolmens Site,” inscription 1501 (2016), criteria (iii) and (iv). <https://whc.unesco.org/en/list/1501/>

⁷⁰UNESCO World Heritage Centre — “Antequera Dolmens Site,” inscription 1501 (2016), criteria (iii) and (iv). <https://whc.unesco.org/en/list/1501/>

⁷¹Antequera Tourism — official monument dossier: the two-line orientation system (Menga→La Peña; Viera→equinox; El Romeral on the axis, → El Torcal). <https://turismo.antequera.es/wp-content/uploads/2011/06/DOLMENES-INGLES.pdf>

⁷²Go-Torcal-Antequera — UNESCO site guide: the four-event astronomical cycle; Viera equinox; El Romeral 199°/winter solstice. <https://www.go-torcal-antequera.com/en/the-el-torcal/unesco-world-heritage-antequera-dolmens-site/>

⁷³Antequera World Heritage — Tholos de El Romeral: corbelled construction, 199° orientation to Camorro de las Siete Mesas, winter-solstice light. <https://www.antequerapatrimoniomundial.com/en/antequera-dolmens/tholos-el-romeral/>

⁷⁴Antequera Tourism — official monument dossier: the two-line orientation system (Menga→La Peña; Viera→equinox; El Romeral on the axis, → El Torcal). <https://turismo.antequera.es/wp-content/uploads/2011/06/DOLMENES-INGLES.pdf>

PART II — THE BUILD

How 32 stones weighing 1,140 tons were cut from a hillside, moved a kilometer, and locked into a machine that has now run for fifty-seven centuries.

CHAPTER 5: THE QUARRY AT CERRO DE LA CRUZ

5.1 Finding where the stones were born

Until recently, nobody knew where Menga's stones came from. The question is harder than it sounds: southern Iberia is all limestone and sandstone, and "the rock is local" had been assumed for a century without proof. In 2023, a team led by geologist José Antonio Lozano Rodríguez published the answer in *Scientific Reports*, after high-resolution geological mapping of a three-kilometer radius around the dolmen, petrographic thin-section analysis of the monument's stones, and comparison against every candidate outcrop [MEASURED]⁷⁵.

The result: the stones of Menga came from **Cerro de la Cruz** — a rocky outcrop roughly 0.85 to 1 kilometer from the dolmen, standing about 50 to 100 meters higher in elevation [MEASURED]⁷⁶⁷⁷. In that same survey, the team recovered hand-made pottery fragments of Neolithic manufacture at the quarry site — modest, but direct, material evidence of activity there in the right period [MEASURED/DOCUMENTED]⁷⁸.

5.2 Five stones for one monument

The petrographic analysis identified five distinct lithologies in the monument, every one matched to specific sedimentary facies at Cerro de la Cruz [MEASURED]⁷⁹:

1. **Bioclastic calcirudite** — a coarse rock of shell and skeletal debris.
2. **Bioclastic calcarenite** — sand-sized carbonate grains with abundant fossil content.
3. **Bioclastic calcirudite/micro-breccia** — a transitional type with small angular fragments and pisoliths (pea-sized concretions), matched specifically to capstones C-2 and C-5 and orthostats O-11 and O-15.
4. **Calcareous breccia** — denser, iron-oxide-rich, and the most porous of the set (porosity approaching 30%).
5. **Bioclastic calcarenite of foreshore origin** — the one type whose precise quarry has not yet been located [OPEN].

All five are sedimentary rocks of Upper Tortonian age (late Miocene, roughly 8–11 million years old), formed in platform and submarine-canyon paleoenvironments — meaning the stone of Menga was born

⁷⁵Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — "The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic," *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁷⁶Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — "The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic," *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁷⁷Archaeology Mag — "Archaeologists uncover advanced engineering in 6,000-year-old Spanish dolmen" (2024). <https://archaeologymag.com/2024/08/archaeologists-uncover-engineering-in-menga-dolmen/>

⁷⁸Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — "The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic," *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁷⁹Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — "The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic," *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

as the floor and canyon walls of a vanished sea [MEASURED]⁸⁰⁸¹. This is not poetic license: the fossil content of the stones — bivalve shells, echinoid burrows, imbricated pebbles dropped by ancient currents — is still visible on the chamber walls, and, as Chapter 14 will show, the builders exploited it deliberately.

5.3 The critical property: softness

Here is the fact on which the entire engineering story pivots. These rocks are, in the vocabulary of modern civil engineering, “**soft stones**” — poorly cemented detrital sediments. The calcarenites show porosities around 13%, the breccias up to nearly 30% [MEASURED]⁸². They are workable, light for their bulk, and — the dangerous flip side — fragile. They chip, spall, and crack under impact and vibration; they drink water by capillarity, gaining weight and losing strength when wet [MEASURED]⁸³.

Every large European megalithic tradition had harder options. Granite, basalt, dolerite, quartzite — Stonehenge’s builders dragged sarsens of far harder stone. Harder rock is heavier per volume and much harder to shape, but it survives transport abuse. The Antequera builders had hard rock close at hand — ophite and dolerite outcrop in the El Torcal zone — and they knew it, because they quarried those rocks *for their tools* [DOCUMENTED]⁸⁴. For the monument itself, they chose soft stone.

This is Flag 5, and Chapter 15 dissects it. For now, record the paradox in its raw form: **the largest stone-moving project in Iberian prehistory was executed in the most fragile class of building stone available** [MEASURED]⁸⁵.

5.4 How the mountain helped: reading the fractures

The quarry study found that Cerro de la Cruz is riddled with systematic tectonic jointing — planes of weakness imposed by the Alpine orogeny. Critically, the joints are subvertical with metric-scale spacing, generating naturally isolated blocks whose dimensions are “compatible with those of the orthostats, capstones and pillars of Menga” [MEASURED]⁸⁶. In the zone designated Quarry #3 (the source of capstones C-2 and C-5), fractures run vertical with penetrative spacing up to 7.5 meters — and capstone C-5 measures $6.95 \times 6 \times 1.88$ meters [MEASURED]⁸⁷. The mountain was, in effect, pre-cut to order.

Even better for the builders: one dominant fracture set dips at about 55° toward the valley and the Menga hill — “obviously a favourable element for their extraction in Neolithic times,” in the study’s own

⁸⁰Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁸¹Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

⁸²Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁸³Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁸⁴Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁸⁵Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁸⁶Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁸⁷Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

words [MEASURED/INFERRED]⁸⁸. A block bounded on its sides by vertical joints and undercut along a valley-dipping plane is a block half-freed by geology.

The extraction method itself can be reconstructed at the level of strong inference [INFERRED]⁸⁹: identify a joint-bounded block; work the joints open with wooden and stone wedges (wood swells when wet — a hydraulic jack made of timber); peck and batter separation surfaces with hammerstones of hard ophite and dolerite; lever the block free onto a prepared bed. Fire-setting — heating rock to induce thermal fracture — is attested in some Iberian quarrying contexts, but there is no definitive evidence for it at Cerro de la Cruz [OPEN].

5.5 A note on what was *not* found

Historical quarrying at Cerro de la Cruz continued into recent centuries and has erased much of the direct Neolithic extraction evidence — tool marks, wedge sockets, spoil heaps [DOCUMENTED]⁹⁰. One “possible discarded megalithic stone” was identified at Quarry #1 [MEASURED]⁹¹. The reconstruction above therefore rests on fracture geometry, block compatibility, lithology matching, and Neolithic pottery — strong circumstantial evidence, openly acknowledged by the researchers. This investigation preserves that honesty: the *where* of the quarry is [MEASURED]; the *how* is [INFERRED] from the fractures, the stones, and experimental analogy.

CHAPTER 6: MOVING MOUNTAINS — THE TRANSPORT PROBLEM

6.1 The worst load in European prehistory

Capstone C-5 weighs an estimated 149.59 ± 9.17 tons — call it 150 [MEASURED]⁹². It is the heaviest stone ever moved as part of the megalithic phenomenon in Iberia, and the second-largest in Europe, after the 335-ton Grand Menhir Brisé [MEASURED]⁹³. But the comparison flatters Brittany and undersells Antequera, for one reason: the Grand Menhir is a *standing stone*, moved and erected once, its journey ending at the ground. C-5 had to be moved and then *placed horizontally atop a structure* — as a load-bearing roof slab of a composite building — without being dropped, twisted, or jarred [MEASURED/INFERRED]⁹⁴. As a component of a composite monument, nothing heavier exists in the European Neolithic record.

And it is soft stone. The transport problem is therefore not merely mass; it is mass $\times$ fragility $\times$ distance

⁸⁸Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁸⁹Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁹⁰Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁹¹Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁹²Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁹³Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁹⁴Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

× slope.

6.2 The route

The route from Cerro de la Cruz to the Menga hill is approximately one kilometer, continuously downhill, on a gentle average slope of about 22° [MEASURED]^{95,96}. Downhill sounds like a gift. For a 150-ton slab of porous calcarenite on a wooden sledge, it is a threat: gravity must be *restrained*, not exploited. A runaway load of that mass is unstoppable, and an impact load of that mass is unrepairable — the slab would shatter, and a shattered C-5 cannot be re-quarried quickly. The failure cost of the transport phase was measured in years of labor [INFERRED].

The transport system the research team reconstructs [INFERRED, consistent with MEASURED geology and with experimental archaeology]:

- **Trackway:** a prepared road of wooden beams or poles laid as sleepers — a low-friction slide-path — leveled and graded the full kilometer. The *Scientific Reports* paper is explicit that “large amounts of wood must have been used... to prepare the roads on which the massive stones were transported” [INFERRED by the authors]⁹⁷.
- **Sledges:** massive timber sleds carrying each block. No rollers are attested, and on a 22° grade rollers would be a liability anyway — a sledge can be braked by friction; a roller train cannot.
- **Rope:** heavy fiber cordage in quantity — for hauling teams, for drag ropes braking the descent, for lashing. The rope requirement of this project, in plant fiber or hide, would itself have been an industrial-scale undertaking: rope is perishable and leaves almost no archaeological trace, which is why the evidence for it is structural — the stones arrived intact — rather than artifactual [INFERRED].
- **Scaffolding:** timber cribbing and platforms in the quarry, for working the joints and loading the sledges [INFERRED]⁹⁸.

6.3 The arithmetic of labor

How many people? No excavation can answer directly, but the physics sets floors. A 150-ton load on a greased wooden sledge over a wooden trackway has a sliding friction coefficient in the range of roughly 0.15–0.3 depending on lubrication; on the flat, that implies a sustained pulling force of tens of tons — hundreds of haulers in harness, or haulers plus capstan-like mechanical advantage through anchored rope turns. Downhill at 22°, the problem inverts into *restraint*: the same hundreds of hands on drag ropes, paying out under control [INFERRED from MEASURED mass and slope].

This investigation adds one logistical observation of its own, following the quarry team’s note that the route runs “always in a downhill direction” [MEASURED]⁹⁹: **the quarry sits 50–100 m above the build site, and the builders had the whole plain to choose from.** The choice of Cerro de la Cruz — correct stone, favorable joints, *and* gravity on the side of the haulers — reads as a single integrated site-selection decision, made before the first block was cut [INFERRED].

⁹⁵MyScience — provenance-study reportage. <https://www.myscience.org/news/2023/>

⁹⁶Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁹⁷Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁹⁸Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

⁹⁹MyScience — provenance-study reportage. <https://www.myscience.org/news/2023/>

6.4 The water hazard en route

One more transport constraint is easy to miss. Porous stone wets easily, and wet porous stone is heavier and weaker [MEASURED]¹⁰⁰. Moving a kilometer means exposure — to rain, to ground moisture, to time. The builders' later obsession with waterproofing the finished monument (Chapter 10) suggests they understood the enemy from the first day at the quarry [INFERRED].

CHAPTER 7: RAISING THE CHAMBER — INSIDE OUT AND UPSIDE DOWN

7.1 The discovery that rewrote the construction sequence

In August 2024, the *Science Advances* paper by Lozano Rodríguez, García Sanjuán, and colleagues — the most detailed geoarchaeological analysis of Menga ever published — proposed a construction sequence that inverted the standard assumption in two senses at once [DOCUMENTED]¹⁰¹.

First inversion: the monument was built from the inside out. The orthostats were not hauled around the outside of an emerging structure and tipped inward from external ramps. They were brought into the interior space of the chamber and set into their sockets *from within* [INFERRED from MEASURED stratigraphic polarity]¹⁰². The proof is geological and elegant, and it is the subject of Chapter 14 in its anomaly context; in brief, the fossil shells, burrows, and pebble imbrications in the stones record which face of each block was the original seabed floor, and that “floor” face consistently ends up *facing into the chamber* — an orientation achievable only by approaching and rotating each stone from inside [MEASURED → INFERRED]¹⁰³.

Second inversion: the chamber was built buried. The builders did not erect a freestanding stone box and then heap a mound over it. They excavated a deep construction trench — a “foundation box” — into the bedrock of the hill, erected the stones within it, and used the surrounding ground itself as the working platform [MEASURED/INFERRED]¹⁰⁴. The consequence is one of the most quietly brilliant features of the entire project: **the capstones never had to be lifted.** They were slid into place at, or near, the original ground level, across the mouths of the trench, and only afterward was the chamber floor excavated down to its final depth inside [INFERRED]¹⁰⁵. A 150-ton roof installed without a single meter of vertical hoisting — the ascending-ramp problem that dominates every discussion of megalithic and pyramid construction simply designed out of existence.

¹⁰⁰Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

¹⁰¹Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹⁰²Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

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¹⁰⁵Archaeology Mag — “Archaeologists uncover advanced engineering in 6,000-year-old Spanish dolmen” (2024). <https://archaeologymag.com/2024/08/archaeologists-uncover-engineering-in-menga-dolmen/>

7.2 The foundation box

The 1991 University of Málaga excavations and the 2005–2006 University of Granada excavations exposed the bases of several orthostats and all three pillars, and the *Science Advances* team reconstructed the full foundation geometry from those records plus every plan, photograph, and rendering made since Mitjana’s 1840s exploration [DOCUMENTED]¹⁰⁶.

The numbers:

- The orthostats nearest the entrance (O-1 and O-23, beneath capstone C-1) are embedded about **2.75 meters** into bedrock [MEASURED]¹⁰⁷.
- The orthostats nearest the back (O-11 and O-13, beneath capstone C-5) are embedded about **3.20 meters** [MEASURED]¹⁰⁸.
- In each case, roughly **one-third of the stone is underground** [MEASURED]¹⁰⁹.
- The foundation ditches form a **stepped terrace descending toward the entrance** [MEASURED]¹¹⁰.
- The ground under the future tumulus was also prepared in stepped terraces, with an external ramp descending toward the backstone — part of the construction logistics [MEASURED]¹¹¹.

The design intent, in the authors’ words, was a “box”: every upright gripped by the earth along a third of its length, base-locked against overturning and sliding [INFERRED by the authors]¹¹². No other gallery dolmen in the Iberian record shows foundation embedment of this depth [DOCUMENTED — comparative absence noted in the literature]¹¹³. Why it matters structurally is the subject of Chapter 16.

7.3 Placing a leaning stone

The orthostats do not stand vertically; they lean inward at a controlled average of 84–85° (Chapter 8). Placing a 20–40-ton slab into a socket at a precise non-vertical angle, and making its dressed side-facets mate with the neighboring slab’s facets, is a problem in controlled instability.

The *Science Advances* team proposes, with explicit experimental support, a **counterweight method** [INFERRED, experimentally tested]¹¹⁴: the upright arrives on its sledge in the interior space, beside its

¹⁰⁶Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹⁰⁷Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

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¹¹¹Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹¹²Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹¹³University of Alcalá repository record for the *Science Advances* paper, including the official abstract (“no precedents in Iberia”). <https://ebuah.uah.es/dspace/handle/10017/63692?show=full>

¹¹⁴Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

socket; a counterweight is attached to shift the stone's center of mass progressively over the socket edge; a wooden wedge between the stone and the socket lip acts simultaneously as fulcrum and as horizontal adjustment screw; as the counterweight is repositioned, the stone rolls gently down into the socket at the designed lean, its side facet pressing against the facet of the stone already placed. The counterweight is then retrieved up the 30° external ramp of the foundation ditch [INFERRED]¹¹⁵. The efficacy of the counterweight placement technique has been demonstrated experimentally by others working on megalith erection [DOCUMENTED]¹¹⁶.

The laser-scan data even preserve the **sequence** of placement. On the left side as one enters, the first upright set was O-10, near the back; then O-9, O-8, O-7, O-6, O-5, O-4, O-3 were placed outward in order, each leaning against the last; a second stage placed the (now missing) O-0 at the entrance with O-1 and O-2 wedged against it; a third stage set O-11 against the backstone [MEASURED geometry → INFERRED sequence]¹¹⁷. The right side has its own sequence. The chamber was assembled like a zipper, from the back outward, each stone leaning on its predecessor — a dependency chain that left no room for error, because an early error propagates down the entire row [INFERRED].

7.4 The pillars

Three pillars survive, standing on the chamber's longitudinal axis beneath the joints of the capstones; the spacing strongly suggests a **fourth pillar, now missing**, originally stood beneath the entranceward roof joint [MEASURED/INFERRED]¹¹⁸. The pillars were socketed as deeply as the orthostats — the OSL dating samples that anchored the construction chronology came from the infill of Pillar 3's own socket [MEASURED]¹¹⁹.

A design note that will matter later: the pillars are not merely props. In the light-architecture of the chamber (Chapter 12), they are also shadow-casting elements that divide the interior light field [INFERRED]¹²⁰. And the well — the deepest anomaly in this book — sits immediately behind Pillar 3, centered on the axis with reference to the backstone and the lateral walls [MEASURED]¹²¹.

¹¹⁵Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹¹⁶Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹¹⁷Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹¹⁸Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹¹⁹García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹²⁰Van den Hoek, P. — “World's Largest Dolmen as Moon- and Sundial,” Hunebed Nieuwscafé (2021): the lunar-standstill dial hypothesis and the R4/R7 step observations. <https://www.hunebednieuwscafe.nl/2021/10/worlds-largest-dolmen-as-moon-and-sundial/>

¹²¹García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

CHAPTER 8: THE GEOMETRY — A CHAMBER OF CONTROLLED ANGLES

8.1 The lean

The laser-scanned survey (TDTEC S.L., 2005 — the first full 3D laser model of any Iberian megalith [DOCUMENTED]¹²²) plus inclinometer measurements give the lean of the orthostats [MEASURED]¹²³:

- **Left side** (as one walks in): average dip $85.2^\circ \pm 1.6^\circ$ from horizontal.
- **Right side**: $84.0^\circ \pm 2.0^\circ$.

The chamber's cross-section is therefore trapezoidal — narrower at the roof than at the floor. The lean is not collapse, not settling, not error: it is consistent across two dozen stones, symmetric in intent, and matched by the dressed angles of the capstones' bearing surfaces [MEASURED]¹²⁴.

Why lean the walls? Three functions stack [INFERRED from engineering first principles, consistent with the authors' analysis]:

1. **Interlock and self-support.** Inward-leaning walls gain stability from the roof load itself: the capstones' weight drives the orthostats' tops inward against each other and against the roof slabs' edges, wedging the system tighter under load rather than prying it apart.
2. **Load path.** The lean begins the outward-and-downward redirection of roof thrust that the capstone arch completes (Chapter 9).
3. **Center of mass.** Deep sockets plus inward lean keep every stone's center of gravity low and inside its base of support — the analysis team's co-author Adolfo Arrieta summarized it as “an intuitive understanding of the benefits of getting the center of gravity of these large stones as low as possible” [DOCUMENTED]¹²⁵.

8.2 The facets: stones that hold hands

The lateral edges of the orthostats are not raw breaks. They are dressed into angled facets, so that neighboring stones meet plane-to-plane and lean on each other across the full height of contact [MEASURED]¹²⁶. The interlock angles measured on the facets run around $77\text{--}80^\circ$ [MEASURED]¹²⁷. Researchers describing the fit reach for the same simile repeatedly: Tetris-like; “the precision... is millimetric” in places [DOCUMENTED]¹²⁸.

From the laser model, the team reconstructed the cutting sequence: each upright's final shape was achieved by a planned series of section removals — trimming the block into a wedge-profiled slab whose two side facets would mate with two different neighbors at two different angles [MEASURED → IN-

¹²²NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

¹²³Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹²⁴Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹²⁵Born to Engineer — “Neolithic Engineers Used Advanced Scientific Principles...” (2024): interlock angles $77\text{--}80^\circ$, Arrieta quotation, plumb-bob interpretation. <https://www.borntoengineer.com/neolithic-engineers-scientific-principles>

¹²⁶Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹²⁷Born to Engineer — “Neolithic Engineers Used Advanced Scientific Principles...” (2024): interlock angles $77\text{--}80^\circ$, Arrieta quotation, plumb-bob interpretation. <https://www.borntoengineer.com/neolithic-engineers-scientific-principles>

¹²⁸Archaeology Mag — “Archaeologists uncover advanced engineering in 6,000-year-old Spanish dolmen” (2024). <https://archaeologymag.com/2024/08/archaeologists-uncover-engineering-in-menga-dolmen/>

FERRED]¹²⁹. This is subtractive manufacture to a tolerance, in soft stone, with stone tools. Each orthostat is a custom part. There are no interchangeable spares in Menga.

8.3 The instruments of precision

How do you hold an 85° lean to $\pm 1.6^\circ$ across two dozen stones with Neolithic technology? The honest answer combines [MEASURED] outcomes with [INFERRED] tools:

- **Plumb line and bob** — a weighted cord gives true vertical; a measured offset from vertical gives any lean angle reproducibly. The research team specifically proposes plumb bobs and triangulation [INFERRED by the authors]¹³⁰.
- **Measuring cords and rods** — proportions transfer perfectly along a tensioned cord.
- **Set-squares or framing tools** in wood — right angles and standard angles are easy to make and check by flipping (any triangle can be validated against its mirror image).
- **A template culture** — the facets' consistency suggests standard angle-gauges applied across the whole build [INFERRED].

No such tools survive — wood and cord perish — and none should be expected to. What survives is the statistical fingerprint of their use: twenty-plus stones holding a mean angle with a standard deviation under two degrees. That standard deviation is the tool's ghost [INFERRED].

8.4 The asymmetry nobody expected

Here the laser model delivered a genuinely strange result, and this investigation flags it as one of the site's most under-discussed anomalies (it enters the Flag Register as part of Flag 2, investigated fully in Chapter 12).

The two sides of the chamber are not equally precise. **The left wall (as one enters) is geometrically disciplined: its plan-view curve fits a third-degree polynomial almost perfectly. The right wall does not — it is comparatively irregular, “fractal,” in the analysts' term** [MEASURED]¹³¹. The contrast extends to weights, volumes, and lithologies: the left side is uniform; the right side is not [MEASURED]¹³².

A cubic-polynomial wall curve is not an accident of rubble masonry. Nor is one-sidedness. Whatever governed the left wall's geometry was deliberate, continuous, and mathematically smooth — and whatever governed the right wall answered to different constraints. The most developed interpretation connects this asymmetry to the solstice light effect (Chapter 12) [INFERRED]¹³³. The raw geometric fact, however, stands on its own, and any future theory of Menga's design must explain not just the curve but its one-sidedness.

¹²⁹Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹³⁰Born to Engineer — “Neolithic Engineers Used Advanced Scientific Principles...” (2024): interlock angles 77–80°, Arrieta quotation, plumb-bob interpretation. <https://www.borntoengineer.com/neolithic-engineers-scientific-principles>

¹³¹NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

¹³²NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

¹³³NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

CHAPTER 9: THE ROOF — FIVE STONES AND AN INVENTION

9.1 The capstones

Five capstones roof the chamber; a sixth, which would have covered the entrance area, is missing — its supporting orthostats survive, half-broken, at the entrance [MEASURED/REPORTED]¹³⁴¹³⁵. The surviving five step upward with the chamber: the interior height rises from about 2.5 m at the entrance to about 3.45 m at the back [MEASURED]¹³⁶.

The capstones are not laid flat. They carry a dip consistent with the overall drainage and load scheme, and capstone C-1 — the entrance capstone — is set visibly *asymmetric* on its supports: it is thicker on one side, and it overhangs the uprights where it is thinner, a distribution that the *Science Advances* team reads as deliberate weight-balancing of an irregular slab [MEASURED → INFERRED]¹³⁷. Even the roof's oddities are engineered.

9.2 C-5: the keystone that predates keystones

And then there is C-5: $6.95 \times 6 \times 1.88$ meters of bioclastic micro-breccia, ~150 tons, the roof of the rear chamber [MEASURED]¹³⁸. Its form was fully revealed by the 1991 University of Málaga excavation into the mound above it, which photographed its complete upper surface [DOCUMENTED]¹³⁹.

C-5's upper surface is not flat. **It is convex — arched in profile.** And the consequence of that curvature is mechanical: a convex-top slab bearing load behaves like a shallow arch or a prestressed beam; it sheds stress outward toward its supports instead of accumulating it at mid-span [MEASURED geometry → engineering INFERENCE]¹⁴⁰¹⁴¹. The research literature describes this as the earliest documented use of a relieving-arch principle in stone architecture — an empirical arch, in the wrong material, three and a half millennia before the Romans made the form canonical [DOCUMENTED/INFERRED]¹⁴².

This investigation pauses here, because this is one of the places where the evidence justifies the strongest language the primary researchers permit themselves, and then some. A relieving arch works because compression follows the curve. To shape a 150-ton soft-stone slab into a shallow arch, you must (a) understand — observationally, not mathematically — that form directs force, (b) design for it before the stone leaves the quarry, since convexity cannot be added in place, and (c) execute the shape in a material that punishes every misplaced blow. All three steps are [INFERRED] from the [MEASURED] geometry. There is no simpler explanation that accounts for the data. An accidental convexity would not need to be

¹³⁴@archeohistories — monument summary thread (X), the original prompt's source for the popular framing; superseded by primary literature throughout this book. <https://x.com/archeohistories/status/1708781792252338569>

¹³⁵Van den Hoek, P. — “World's Largest Dolmen as Moon- and Sundial,” Hunebed Nieuwscafé (2021): the lunar-standstill dial hypothesis and the R4/R7 step observations. <https://www.hunebednieuwscafe.nl/2021/10/worlds-largest-dolmen-as-moon-and-sundial/>

¹³⁶Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹³⁷Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹³⁸Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

¹³⁹Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

¹⁴⁰Archaeology Mag — “Archaeologists uncover advanced engineering in 6,000-year-old Spanish dolmen” (2024). <https://archaeologymag.com/2024/08/archaeologists-uncover-engineering-in-menga-dolmen/>

¹⁴¹Born to Engineer — “Neolithic Engineers Used Advanced Scientific Principles...” (2024): interlock angles 77–80°, Arrieta quotation, plumb-bob interpretation. <https://www.born-to-engineer.com/neolithic-engineers-scientific-principles>

¹⁴²Archaeology Mag — “Archaeologists uncover advanced engineering in 6,000-year-old Spanish dolmen” (2024). <https://archaeologymag.com/2024/08/archaeologists-uncover-engineering-in-menga-dolmen/>

dressed; a decorative convexity serves no visible purpose from inside the chamber — the curve is on the *top* of the stone, hidden under the mound, visible only to the builders and to a 1991 excavation camera.

9.3 The pillar-and-joint system

The capstones meet over the pillars: each roof joint is supported by a pillar on the axis [MEASURED]¹⁴³. The system is redundant — each capstone is carried by the two flanking walls *and* by the pillar line; the loss of any single support does not collapse the roof. Redundancy of this kind is the signature of builders who expected the structure to be attacked — by time, by water, by earthquake — and meant it to survive [INFERRED].

9.4 The no-ramp solution, restated

It is worth restating the construction trick of Chapter 7 in roof terms, because it is the answer to the question every visitor asks: *how did they get 150 tons up there?* They didn't. **There was no “up.”** The chamber was built in a trench; the capstones were slid across at ground level; the interior was then excavated down to its final floor beneath the already-placed roof [INFERRED by the research team, from MEASURED foundation and trench geometry]¹⁴⁴¹⁴⁵. The dolmen was not raised out of the hill so much as *revealed inside* it — carved out from under its own roof like a ship in a bottle, in reverse.

The sophistication of this inversion is easy to miss because it leaves no artifact: it is a decision, visible only in the stratigraphy. It is also, as far as the comparative record shows, unique at this scale and date [DOCUMENTED]¹⁴⁶.

CHAPTER 10: THE ARMOR — ANATOMY OF THE TUMULUS

10.1 Not a pile of dirt

The mound that swallows the chamber — the “hill that is not a hill” — measures about 48.3 m (NE–SW) by 44 m (NW–SE), covering roughly 2,730 m² [MEASURED]¹⁴⁷. The 2005–2006 Granada excavation cut a 27.4-meter trench across its northwestern quadrant and read its structure like a tree core [DOCUMENTED]¹⁴⁸:

- **Basal layer** (~1.5 m thick): very dark soil rich in pottery, flint, and organics — the pre-monument activity surface of the hill, sealed in place. Eight of the radiocarbon dates that anchor the construction window came from this layer [MEASURED]¹⁴⁹.

¹⁴³Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

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¹⁴⁶University of Alcalá repository record for the *Science Advances* paper, including the official abstract (“no precedents in Iberia”). <https://ebuah.uah.es/dspace/handle/10017/63692?show=full>

¹⁴⁷García Sanjuán, L., et al. — author manuscript of the multimethod dating paper (University of Southampton ePrints), including mound dimensions and the catalog of historical radiocarbon determinations. https://eprints.soton.ac.uk/470780/1/Garcia_Sanjuan_et_al_2022_A_multimethod_approach_to_the_genesis_of_menga_a_world_heritage_megalith.pdf

¹⁴⁸García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁴⁹García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar

- **Upper mound** (~3 m thick): the engineered body — **alternating courses of carefully locked medium-size sandstone slabs and rammed clay** [MEASURED]¹⁵⁰.

Read that upper structure as an engineer: interleaved slab-and-clay courses are a composite armor. The stone courses give compressive strength, shed and interrupt water percolation, and lock against erosion gullies; the rammed clay courses seal capillary paths and bind the mass. It is, in effect, a waterproof shell over the soft-stone chamber — and the quarry study concludes bluntly that the tumulus was made waterproof *because* the builders knew their stone drank water and weakened [MEASURED structure → INFERRED function]¹⁵¹. “Undoubtedly, this robust device... has played a major part in its preservation, protecting it from weathering and geological events,” the dating team writes [DOCUMENTED]¹⁵².

10.2 The mound as structure

The tumulus is also part of the machine’s statics. Its mass buttresses the chamber externally, and — as the fracture of orthostat O-14 shows (Chapter 16) — its load was factored, sometimes imperfectly, into the original calculations [MEASURED]¹⁵³. One radiocarbon determination from the mound’s western edge is later than the rest, and the excavators interpret it as a probable **expansion of the tumulus** sometime after primary construction — the monument was maintained and enlarged, not abandoned [MEASURED/INFERRED]¹⁵⁴.

10.3 Reading the whole build as one mind

Part II closes with a synthesis this investigation offers as its own assessment [INFERRED from the accumulated MEASURED record]:

Every major decision in the build — soft stone from a jointed quarry one kilometer uphill; downhill transport on timber; a foundation box three meters into bedrock; inside-out erection with counterweights; constant-lean walls with interlocking facets; a pillar-redundant roof set without lifting; a hidden relieving arch; a waterproof armored mound — is the *lowest-risk solution available with stone-age materials to the specific problem at hand*. Whoever designed Menga was not decorating. They were de-risking eternity. The monument is best read not as a tomb with engineering features, but as an engineering project that happened to be a tomb — a distinction that becomes the fulcrum of Part III.

End of Part II. Part III takes the ten flags one by one.

3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁵⁰García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁵¹Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

¹⁵²García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁵³Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹⁵⁴García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

PART III — THE FLAGS

Ten documented anomalies. Each is stated as measured, investigated to the bottom of the evidence, and left exactly where the evidence leaves it — no further.

CHAPTER 11: FLAG 1 — THE 45-DEGREE ANOMALY

FLAG STATEMENT. *The central axis of Menga is oriented to azimuth $\sim 45^\circ$ (northeast) — outside the entire annual range of sunrise, in a cultural dataset where 97% of measured tombs face the rising or climbing sun [MEASURED]¹⁵⁵¹⁵⁶.*

11.1 How strange is 45° , exactly?

Precision matters here. At Antequera's latitude ($\sim 37^\circ\text{N}$), the sun rises at its northernmost point — summer solstice — at an azimuth of roughly 59° – 60° , varying with horizon altitude. The statistical sunrise arc Hoskin measured for Iberian tombs runs 60° – 130° [MEASURED]¹⁵⁷. Menga's 45° axis sits about 10° – 15° beyond even the solstitial extreme [MEASURED]¹⁵⁸. No sunrise, on any day of any year, occurs down Menga's corridor.

In Hoskin's western Iberian sample of 334 tombs, 324 obeyed the solar rule [MEASURED]¹⁵⁹. The rule held for a thousand years before Menga (Arroyo Saladillo's horizon) and for two thousand years after. The one tomb that breaks it is the earliest of the great dolmens, the largest, and the most carefully engineered. Rule-breaking at this level of statistical discipline is not noise. It is signal [INFERRED].

11.2 What the corridor actually points at

Extend Menga's axis northeast across the plain. It strikes the northern face of La Peña de los Enamorados — and not the mountain in general: it strikes the plane of the 100-meter cliff forming the sleeping figure's *chin*, the Matacabras sector, where the rock-art sanctuary already stood before 3800 BCE [MEASURED]¹⁶⁰. The Quaternary Research dating paper states the geometry without equivocation: “Menga faces the northern sector of La Peña de los Enamorados mountain, where at the time of its construction intense activity was already taking place around the Matacabras sanctuary” [MEASURED/INFERRED]¹⁶¹.

¹⁵⁵García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁵⁶Hoskin, M. — “Orientations of dolmens of Western Europe,” *Complutum* 20(2), 165–175 (2009). The statistical law of Iberian tomb orientations (97% SR; 96.4% SR/SC). <https://revistas.ucm.es/index.php/CMPL/article/download/CMPL0909220165A/29024/30185>

¹⁵⁷Hoskin, M. — “Orientations of dolmens of Western Europe,” *Complutum* 20(2), 165–175 (2009). The statistical law of Iberian tomb orientations (97% SR; 96.4% SR/SC). <https://revistas.ucm.es/index.php/CMPL/article/download/CMPL0909220165A/29024/30185>

¹⁵⁸García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

¹⁵⁹Hoskin, M. — “Orientations of dolmens of Western Europe,” *Complutum* 20(2), 165–175 (2009). The statistical law of Iberian tomb orientations (97% SR; 96.4% SR/SC). <https://revistas.ucm.es/index.php/CMPL/article/download/CMPL0909220165A/29024/30185>

¹⁶⁰García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña's geology and the 45° alignment's cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

¹⁶¹García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar

Michael Hoskin himself — the man whose dataset defines the rule Menga breaks — confirmed in 2002 that Menga’s orientation is terrestrial, targeting the mountain [DOCUMENTED]¹⁶². The visual relationship was noticed as early as 1851 by Lady Louisa Tenison, who recorded the view of La Peña framed by the dolmen’s entrance [DOCUMENTED]¹⁶³.

11.3 What the break purchased

Breaking the solar rule was not free. A tomb oriented inside the sunrise arc gets its ritual light *through the door*, down the corridor, for free — that is how Viera (equinox sunrise) and thousands of other tombs work [MEASURED]¹⁶⁴. Menga’s builders, by aiming at the mountain, forfeited the corridor-light mechanism and had to replace it with something else. Chapter 12 shows what they built instead: an indirect, wall-mediated solstice light effect that required reshaping the chamber’s left wall into a polynomial curve [MEASURED]¹⁶⁵.

In other words: **the builders accepted an enormous geometric penalty to keep the mountain in the crosshairs, then spent additional engineering to recover a solar effect within the constraint** [INFERRED from MEASURED geometry]. The mountain outranked the sun. Whatever La Peña represented — ancestor, sleeper, guardian, landmark of the crossroads — it was, at the moment of decision, the single most important thing in their world [INFERRED; the referent’s identity is OPEN].

11.4 Alternative explanations, examined and weighed

- **“The orientation is random / dictated by terrain.”** Weak. The hill offers a 360° choice; the quarry approach direction does not fix the chamber’s internal axis; and the axis hits a specific cliff plane, not a generic horizon [MEASURED]¹⁶⁶.
- **“It targets the summer solstice after all.”** Common in popular sources, geometrically wrong for the *axis* — but with an important kernel of truth: the *chamber design* harvests solstice light without an axial alignment (Chapter 12) [MEASURED]¹⁶⁷¹⁶⁸.
- **“It targets a lunar event.”** Proposed independently by at least one archaeoastronomer working from the plan geometry (the right wall as a lunar-standstill dial) [REPORTED/INFERRED — non-peer-reviewed]¹⁶⁹. The lunar claims are examined in Chapter 12; they are consistent with some plan features but remain unverified by peer-reviewed measurement [OPEN].

3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁶²Antequera Tourism — official monument dossier: the two-line orientation system (Menga→La Peña; Viera→equinox; El Romeral on the axis, → El Torcal). <https://turismo.antequera.es/wp-content/uploads/2011/06/DOLMENES-INGLES.pdf>

¹⁶³Museos de Andalucía — official site brochure (English): research history from 1530 onward, El Toro sequence, settlement catalog, Hoskin’s 2002 confirmations. <https://www.museosdeandalucia.es/documents/1971642/0/Folieto+en+ingl%C3%A9s/>

¹⁶⁴Antequera Tourism — official monument dossier: the two-line orientation system (Menga→La Peña; Viera→equinox; El Romeral on the axis, → El Torcal). <https://turismo.antequera.es/wp-content/uploads/2011/06/DOLMENES-INGLES.pdf>

¹⁶⁵NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

¹⁶⁶García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

¹⁶⁷García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁶⁸NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

¹⁶⁹Van den Hoek, P. — “World’s Largest Dolmen as Moon- and Sundial,” Hunebed Nieuwscafé (2021): the lunar-standstill dial hypothesis and the R4/R7 step observations. <https://www.hunebednieuwscafe.nl/2021/10/worlds-largest-dolmen-as-moon-and-sundial/>

- **“It targets Matacabras specifically.”** The strongest terrestrial reading, and the one the dating team favors; the sanctuary predates the monument and sits exactly on the axis [MEASURED]¹⁷⁰. This investigation notes one refinement: axis and sanctuary alignment are compatible, but the axis also strikes the *cliff plane itself* — the chin of the sleeping figure — and the two targets (sanctuary and chin) are not separable at the precision of the survey [MEASURED → the distinction remains OPEN].

11.5 Finding

Flag 1 resolves, at the level of evidence, to this: **the orientation anomaly is fully explained as a deliberate, prioritized alignment on La Peña de los Enamorados’ northern face, overriding the pan-Iberian solar custom — the only such case in continental Europe** [MEASURED]¹⁷¹. What La Peña *meant* — why a mountain outweighed the sun — is beyond the reach of measurement [OPEN]. The clue chain stops at the cliff face. We follow it there, and no further.

CHAPTER 12: FLAG 2 — THE LIGHT MACHINE

FLAG STATEMENT. *Although Menga’s axis lies outside the solar arc, its interior geometry produces a precise summer-solstice light effect: the left side of the chamber remains in shadow while much of the right side is illuminated — an effect achieved, in part, by shaping the left wall’s curve to a third-degree polynomial* [MEASURED]¹⁷²¹⁷³.

12.1 The effect itself

On mornings around the summer solstice, the rising sun — appearing far to the right of the corridor axis, since the corridor points 10–15° north of it — sends light through the entrance at an oblique angle. What happens inside has been documented by direct observation and modeling [DOCUMENTED/MEASURED]¹⁷⁴¹⁷⁵:

- The light does not strike the backstone (as it would in a conventionally aligned passage grave).
- Instead, it sweeps along the **right-hand wall**, illuminating a broad band of the chamber’s right side while the **left side remains in shadow**.
- The boundary between light and shadow falls along the chamber’s longitudinal structure, in the zone of the pillars.

The dating team’s summary is careful and precise: “although Menga’s orientation prioritized a land

¹⁷⁰García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁷¹UNESCO World Heritage Centre — “Antequera Dolmens Site,” inscription 1501 (2016), criteria (iii) and (iv). <https://whc.unesco.org/en/list/1501/>

¹⁷²NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

¹⁷³García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁷⁴NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

¹⁷⁵Casa Olea — solstice observation guide (documenting the public observation of Menga’s solstice light). <https://www.dev.casaolea.com/blog/best-places-see-summer-solstice-andalucia>

‘target’... its design also sought to integrate architecture and sunlight... thus seeking a sunlight effect in a manner best reflected in Newgrange (Ireland)” [INFERRED by the authors from MEASURED geometry]¹⁷⁶. The Newgrange comparison is structural, not decorative: at Newgrange, a tomb deliberately engineered outside any useful orientation nonetheless harvests a razor-precise solstice event through a specially built aperture (the roof box). At Menga, the aperture is the entrance and the engineered element is the wall geometry [INFERRED].

12.2 The polynomial wall

Return now to the asymmetry measured in Chapter 8: the left wall’s plan fits a third-degree polynomial; the right wall is irregular [MEASURED]¹⁷⁷. The geometric analysts’ interpretation is that the left wall’s smooth curve functions as the **shadow-generating surface** — its continuous curvature controls exactly where the penumbra falls at solstice dawn — while the right wall is the *receiving* surface, the illuminated screen [INFERRED]¹⁷⁸.

If that reading is correct, the chamber is a two-part optical instrument: one wall engineered to *cast* a shaped shadow, one wall left as a bright field, with the axis of symmetry — and the pillar line — as the divider. The whole interior is a dial. And note the cost accounting from Chapter 11: this instrument is what the builders bought with their 45° rule-break. They gave up the free corridor-alignment mechanism that every other Iberian tomb used, and built a harder one, because the corridor had to point at the mountain [INFERRED].

This investigation marks the optical interpretation as [INFERRED] — plausible, geometrically supported, not provable — but underlines the raw data beneath it, which are [MEASURED] and not in dispute: the solstice light asymmetry is real, observed, and reproducible; the left wall’s polynomial fit is real; the contrast between the two walls is real.

12.3 The lunar claims — investigated, not adopted

An independent Dutch archaeoastronomer, Peter van den Hoek, has proposed a more elaborate reading of the right wall as a combined **sun-and-moon dial** [REPORTED — published in a non-peer-reviewed venue]¹⁷⁹. In his model:

- At midsummer sunrise, the shadow boundary falls at the joint between orthostats R4 and R5 — and R4 visibly “jumps forward” in the plan, as if repositioned to mark the line.
- At major lunar standstill (the moon’s extreme northern rise, recurring every 18.6 years), rising moonlight would illuminate the entire right wall, with the shadow beginning exactly at the back-stone/R1 joint.
- At minor lunar standstill, moonlight would reach only to the R7/R8 joint — and R7, like R4, steps forward in the plan.
- The standstill cycle would thus be readable off the wall, and with it eclipse seasons.

¹⁷⁶García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁷⁷NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

¹⁷⁸NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

¹⁷⁹Van den Hoek, P. — “World’s Largest Dolmen as Moon- and Sundial,” Hunebed Nieuwscafé (2021): the lunar-standstill dial hypothesis and the R4/R7 step observations. <https://www.hunebednieuwscafe.nl/2021/10/worlds-largest-dolmen-as-moon-and-sundial/>

What should an evidence-led investigation do with this? Three things.

First, acknowledge the [MEASURED] kernel: orthostats R4 and R7 *do* step forward in the plan [MEASURED — visible in the published laser-scan plans]¹⁸⁰¹⁸¹. Second, apply the test of provenance: the solar effect at the R4/R5 joint has independent observational support; the full lunar scheme has not, to this investigation's knowledge, been subjected to peer-reviewed survey with horizon-altitude measurement — and moonlight-on-wall claims are notoriously sensitive to the missing entrance capstone, which changes the light aperture [INFERRED critique]. Third, note the structural plausibility: a culture tracking a 9.3/18.6-year lunar cycle across generations leaves no other trace; architecture is exactly where such knowledge could hide [INFERRED].

Verdict: the solar instrument is [MEASURED/DOCUMENTED]; the lunar extension is [OPEN] — neither established nor dismissible, pending survey-grade measurement.

12.4 The deeper pattern: light as a regional language

Menga's light effect is not an isolated trick; it is the founding document of a regional grammar [DOCUMENTED]¹⁸²¹⁸³:

- **Viera** (built after Menga): corridor into the equinoctial sunrise arc [MEASURED]¹⁸⁴.
- **Piedras Blancas** (in the giant's chest): triangular stones funneling solstice dawn light to the chamber back [MEASURED]¹⁸⁵.
- **El Romeral** (a millennium later): corridor to 199° southwest — one of the very few west-facing orientations in Iberia — aimed at El Torcal's Camorro de las Siete Mesas, with winter-solstice sunlight penetrating to the second chamber [MEASURED]¹⁸⁶¹⁸⁷.

Four monuments, four different solutions to the same problem: how to bind the sky to the stone. At Menga the solution is the most complex because the constraint was the hardest — *keep the mountain, keep the sun*. The builders refused to choose [INFERRED].

12.5 Finding

Flag 2 resolves to: **a deliberate, engineered solstice light effect compensating for the non-solar axis, executed through asymmetric wall geometry of measured mathematical regularity** [MEASURED geometry; INFERRED instrument-function]. The one-sidedness of the precision — the polynomial left wall against the fractal right — is the strongest physical evidence that the chamber was

¹⁸⁰Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

¹⁸¹Van den Hoek, P. — “World's Largest Dolmen as Moon- and Sundial,” Hunebed Nieuwscafé (2021): the lunar-standstill dial hypothesis and the R4/R7 step observations. <https://www.hunebednieuwscafe.nl/2021/10/worlds-largest-dolmen-as-moon-and-sundial/>

¹⁸²García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña's geology and the 45° alignment's cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

¹⁸³Antequera Tourism — official monument dossier: the two-line orientation system (Menga→La Peña; Viera→equinox; El Romeral on the axis, → El Torcal). <https://turismo.antequera.es/wp-content/uploads/2011/06/DOLMENES-INGLES.pdf>

¹⁸⁴Antequera Tourism — official monument dossier: the two-line orientation system (Menga→La Peña; Viera→equinox; El Romeral on the axis, → El Torcal). <https://turismo.antequera.es/wp-content/uploads/2011/06/DOLMENES-INGLES.pdf>

¹⁸⁵HeritageDaily — Piedras Blancas reportage: ripple-marked slabs, solstice-funnel stones. <https://www.heritagedaily.com/2023/05/complex-funary-monument-found-at-foot-of-la-pena-de-los-enamorados/147231>

¹⁸⁶Go-Torcal-Antequera — UNESCO site guide: the four-event astronomical cycle; Viera equinox; El Romeral 199°/winter solstice. <https://www.go-torcal-antequera.com/en/the-el-torcal/unesco-world-heritage-antequera-dolmens-site/>

¹⁸⁷Antequera World Heritage — Tholos de El Romeral: corbelled construction, 199° orientation to Camorro de las Siete Mesas, winter-solstice light. <https://www.antequerapatrimoniomundial.com/en/antequera-dolmens/tholos-el-romeral/>

conceived as an optical device and not merely a container [MEASURED]¹⁸⁸. The lunar-dial hypothesis remains [OPEN].

CHAPTER 13: FLAG 3 — THE WELL

FLAG STATEMENT. *Inside the rear of the chamber, centered on the longitudinal axis immediately behind Pillar 3, a cylindrical shaft 1.5 m in diameter descends 19.4 m through bedrock to the aquifer. No other European megalithic chamber of the period contains a comparable engineered water feature [MEASURED]¹⁸⁹.*

This is the deepest anomaly in the investigation — literally and in terms of what it implies. Give it the slowest walk-through.

13.1 The object

The shaft is circular, about 1.5 meters across at the mouth, narrowing slightly toward the bottom, cut with a smooth, geometric finish through the calcarenite bedrock to a depth of 19.4 meters, where it reaches the groundwater of the Antequera plain [MEASURED]¹⁹⁰. It holds water today; sampling in 2017 confirmed it is fresh, potable groundwater [MEASURED]¹⁹¹.

Its position is architecturally exact: on the chamber’s axis of symmetry, in the rear third of the chamber (where the chamber reaches its maximum width of about 6 m), just behind the third pillar [MEASURED]^{192,193}. The dating paper lists it first among the elements that make Menga unique: “a unique hydraulic feature within the context of the megalithic phenomenon worldwide” [DOCUMENTED]¹⁹⁴.

One geometric relationship, noted by observers, deserves a line of its own: **the well’s depth (19.4 m) is very nearly equal to the length of the chambered gallery** [MEASURED → noted as correspondence]¹⁹⁵. Proportional rhymes of this kind are exactly what a measuring-cord culture produces; whether this one is design or coincidence is [OPEN].

¹⁸⁸NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

¹⁸⁹García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁹⁰García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁹¹García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well’s excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

¹⁹²@archeohistories — monument summary thread (X), the original prompt’s source for the popular framing; superseded by primary literature throughout this book. <https://x.com/archeohistories/status/1708781792252338569>

¹⁹³García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁹⁴García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

¹⁹⁵Go-Torcal-Antequera — “Where a Medieval Legend Explains a 19-Metre-Deep Shaft”: the Menga/Dominga etymology, the 1530 document, Rojas 1874, the leper and sorceress variants. <https://www.go-torcal-antequera.com/en/magazine/legend-woman-menga-dolmen/>

13.2 The water problem, revisited

Chapter 1 planted the fact; now it detonates. The Antequera region is a landscape of brackish springs and salt lakes — gypsum- and halite-laden Triassic ground delivering sulfate- and chloride-rich water unfit for regular consumption [DOCUMENTED]¹⁹⁶. Fresh water is localized and precious. The Menga well taps fresh water, in a region where that is the exception [MEASURED]¹⁹⁷.

The research team that excavated and studied the well frames the implication: the basic pattern of human interaction with the region's water resources “was already laid out in the Late Neolithic” — and they explicitly link the accumulation of “wealth, technical know-how and social capital that led to the construction of such a magnificent monument” to control of this hydraulic niche [INFERRED by the authors]¹⁹⁸. In plain terms: **whoever held the fresh water on a brackish plain held the plain**. A monument that owns its own well owns more than a tomb [INFERRED].

13.3 Was it dug with the dolmen? The honest state of the debate

Here this investigation must be scrupulous, because the popular record blurts “5,700-year-old well” while the scientific record is more careful.

What is [MEASURED]:

- The shaft exists, is artificial, is geometrically finished, and is integrated into the chamber's symmetry [MEASURED].
- Its infill has been radiocarbon dated by sixteen determinations — *all of them* historical, falling between the 15th and 18th centuries CE [MEASURED]. The fill is not prehistoric.
- The well's mouth sits under the tumulus and the structure; nothing about the chamber's fabric suggests the shaft was cut through finished architecture later [MEASURED/INFERRED — no cut-through disturbance documented].

What is argued [INFERRED by the excavators]:

- The well is architecturally integrated — centered, axially aligned, positioned relative to Pillar 3 and the backstone with the same discipline as every other element — which suggests design intent from the beginning [INFERRED].
- Its depth matches the aquifer, which the builders demonstrably knew how to find [INFERRED].
- Comparative context: engineered wells of Neolithic date exist elsewhere in the Old World — the famous early Neolithic wells of the eastern Mediterranean (e.g., Kissonerga-Mylouthkia, Cyprus) — so the capability is not anachronistic [DOCUMENTED].

What is unknown [OPEN]:

- No datable material from the well's *original* construction or prehistoric use has ever been recovered. The shaft was cleared to its bottom before or during its early-1700s backfilling, so the primary fill — which would have answered the question — is gone [MEASURED — absence documented].

¹⁹⁶García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well's excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

¹⁹⁷García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well's excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

¹⁹⁸García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well's excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

This investigation's finding: the prehistoric origin of the well is the **best-supported reading** — architectural integration is strong circumstantial evidence, and digging a 19.4-meter shaft inside a roofed ancient chamber at any later date would be a bizarre, dangerous, undocumented operation — but it remains a reading, not a measurement. Honest tag: [INFERRED, strongly; formally OPEN].

13.4 The well's biography — what the fill told us

The 2005 excavation of the shaft (led by Francisco Carrión of the University of Granada, with David García González descending into the bore) excavated the full 19.4 m and produced a forensic biography of the well's last three centuries [DOCUMENTED/MEASURED]¹⁹⁹²⁰⁰:

- The shaft had been **completely backfilled in the early 18th century** — an operation requiring roughly 50 tons of material to be hauled into the chamber and dumped down the bore [MEASURED/INFERRED]²⁰¹.
- The fill was heterogeneous: local mound material, but also material brought from a distance — **including complete carcasses of dead animals** (cow, horse, donkey, dog, cat, hare; some articulated, i.e., thrown in whole) [MEASURED]²⁰².
- The effort and the animal carcasses together read as an **official, deliberate suppression** — not casual silting, not a farmer filling a hazard, but an organized act to decommission the well and discourage return [INFERRED]²⁰³.
- Context for *why*: late-16th-century references describe the “Cueva de Menga” as a place of pagan practices and “demonic rituals”; pottery water vessels around the site attest to people coming for the water; the legend of the leper woman (Chapter 22) records belief in the water's healing power [REPORTED/DOCUMENTED]²⁰⁴²⁰⁵. Several crosses are carved into an orthostat near the entrance — an effort to sanctify, or seal, a place the Church regarded as unchristian [MEASURED/REPORTED]²⁰⁶.
- After the filling, the well was *forgotten*. Mitjana, excavating in the 1840s, dug into the shaft's top, found nothing, and concluded — fatefully for a century of scholarship — that the feature was not significant [DOCUMENTED]²⁰⁷. Lady Louisa Tenison's 1853 book records the open shaft (“five feet wide and forty feet deep”) — the last written notice of it [DOCUMENTED]²⁰⁸. Between

¹⁹⁹García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well's excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

²⁰⁰Andalucia.com — “Dolmens of Antequera”: Mitjana's 1842–47 dig and shaft note; Tenison 1852/1853 warning; Carrión's 2005 well excavation; 19th-century visitor list. <https://www.andalucia.com/antequera/dolmens-de-menga.htm>

²⁰¹Hobthrush — “Dolmen de Menga”: the well's early-1700s suppression, ~50 t of fill, animal carcasses, carved crosses, and the pre-suppression clearing hypothesis (tagged [REPORTED]/[INFERRED] in the main text). <https://hobthrush.co.uk/dolmen-de-menga-antequera-spain/>

²⁰²García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well's excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

²⁰³Hobthrush — “Dolmen de Menga”: the well's early-1700s suppression, ~50 t of fill, animal carcasses, carved crosses, and the pre-suppression clearing hypothesis (tagged [REPORTED]/[INFERRED] in the main text). <https://hobthrush.co.uk/dolmen-de-menga-antequera-spain/>

²⁰⁴Hobthrush — “Dolmen de Menga”: the well's early-1700s suppression, ~50 t of fill, animal carcasses, carved crosses, and the pre-suppression clearing hypothesis (tagged [REPORTED]/[INFERRED] in the main text). <https://hobthrush.co.uk/dolmen-de-menga-antequera-spain/>

²⁰⁵Go-Torcal-Antequera — “Where a Medieval Legend Explains a 19-Metre-Deep Shaft”: the Menga/Dominga etymology, the 1530 document, Rojas 1874, the leper and sorceress variants. <https://www.go-torcal-antequera.com/en/magazine/legend-woman-menga-dolmen/>

²⁰⁶Hobthrush — “Dolmen de Menga”: the well's early-1700s suppression, ~50 t of fill, animal carcasses, carved crosses, and the pre-suppression clearing hypothesis (tagged [REPORTED]/[INFERRED] in the main text). <https://hobthrush.co.uk/dolmen-de-menga-antequera-spain/>

²⁰⁷Andalucia.com — “Dolmens of Antequera”: Mitjana's 1842–47 dig and shaft note; Tenison 1852/1853 warning; Carrión's 2005 well excavation; 19th-century visitor list. <https://www.andalucia.com/antequera/dolmens-de-menga.htm>

²⁰⁸Museos de Andalucía — official site brochure (English): research history from 1530 onward, El Toro sequence, settlement catalog, Hoskin's 2002 confirmations. <https://www.museosdeandalucia.es/documents/1971642/0/Fol>

roughly 1850 and 1868 it was re-backfilled, and **no publication mentioned it again until its rediscovery in 2005** [DOCUMENTED]²⁰⁹.

A well so troublesome that authority spent 50 tons of fill and a cartload of dead animals to erase it — and so forgettable that it then vanished from the record for 150 years — is not a utilitarian hole. It was, in the early modern period, the living heart of whatever gave the “Cueva de Menga” its reputation [INFERRED]. And its legend-twin — the healing water sought by the leper woman — preserves, in folk form, exactly the function (fresh, special, perhaps sacred water) that the hydrogeology makes plausible [REPORTED legend MEASURED water chemistry; the connection is INFERRED].

13.5 Comparative uniqueness

To be precise about the flag: wells and water shafts of prehistoric date exist in Europe and the Near East [DOCUMENTED]²¹⁰; sacred wells and *pozzi sacri* become common in Bronze Age Sardinia — a millennium and more later [DOCUMENTED]²¹¹. What does not exist, anywhere in the megalithic record of Europe at this date, is **a deep engineered water shaft integrated into the chamber of a megalithic monument** [DOCUMENTED]²¹². The well is a class of one.

13.6 Finding

Flag 3 resolves to: **a fresh-water well of unique design and integration, almost certainly original to the monument, giving the structure a hydraulic function unparalleled in megalithic Europe — and a continuous biography of water-use, suppression, and forgetting that runs from the Neolithic to the 18th century** [MEASURED/INFERRED as detailed above]. Whether the well’s primordial role was practical (water for the community), ritual (water for the dead or the rites), or both, is [OPEN]. This investigation notes, without adopting, that on a brackish plain the distinction may have meant nothing to the builders: where fresh water is rare, water *is* the sacred [INFERRED].

CHAPTER 14: FLAG 4 — THE INSIDE-OUT STONES

FLAG STATEMENT. *Every orthostat and structural element of Menga was placed with the same geological “base” — the original lower face of the sedimentary bed it was cut from — oriented toward the interior of the chamber. Fossil shells, burrows, and pebble imbrications all face inward. Every block was deliberately rotated from its natural quarry orientation* [MEASURED]²¹³.

leto+en+ingl%C3%A9s/

²⁰⁹García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well’s excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

²¹⁰García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well’s excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

²¹¹García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well’s excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

²¹²García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²¹³Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

14.1 How do you read which way is “up” in a stone?

Sedimentary rock remembers its birth. When the *Science Advances* team examined Menga’s stones, they found three independent polarity indicators preserved in the rock [MEASURED]²¹⁴:

- **Bivalve shells:** disarticulated shells lie concave-side-up on the ancient seafloor; in Menga’s orthostats, the concave faces point overwhelmingly toward the chamber interior.
- **Echinoid bioturbation:** burrows left by sea-urchin-like creatures have flat bases marking the original bed floor — in upright O-14, the burrow floor faces inward.
- **Imbricated pebbles:** pebbles stacked by ancient currents record flow direction and bed orientation — in O-22, they face inward.

And the clincher: the quarry strata at Cerro de la Cruz lie **horizontal** today, as they did when quarried [MEASURED]²¹⁵. So the stones were not flipped by geology; they were flipped — rotated — *by hand*, one by one, every single one, so that the old seafloor face of the rock looks into the chamber.

14.2 What the rotation proves about the build

The polarity pattern is the key evidence for the inside-out erection sequence of Chapter 7: stones approached and rotated from within the chamber land in exactly this configuration, and it is hard to achieve otherwise [MEASURED → INFERRED]²¹⁶. So one part of Flag 4 is simply forensic: it tells us *how* they built.

But the consistency — *every* stone, not most — exceeds what construction technique alone requires. The technique explains why inward-facing orientation was *possible*; it does not explain why it was *universal* [INFERRED].

14.3 Why face the fossils inward? Three readings

- **The functional reading [INFERRED, offered by the research literature]:** the bed’s natural “roof” (top) face is the weathering-resistant skin of the rock; orienting it outward presents the tougher face to the mound’s moisture and pressure, protecting the stone — and hides the more erodible floor-face inside, where the chamber air is stable. On this reading, the polarity is engineering: stone rotated for durability ²¹⁷.
- **The aesthetic/symbolic reading [INFERRED]:** the floor-faces carry the visible fossils — shells, burrows, ripples of the ancient sea. The builders of Piedras Blancas demonstrably *chose* stones for their marine markings, placing ripple-textured slabs at the most visible positions, a choice echoed at El Romeral [MEASURED]²¹⁸. On this reading, Menga’s walls are a gallery of the sea: the monument’s interior lined with the memory of the water world the stone came from — fitting company for a chamber with a well at its heart.

²¹⁴Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

²¹⁵Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

²¹⁶Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

²¹⁷Born to Engineer — “Neolithic Engineers Used Advanced Scientific Principles...” (2024): interlock angles 77–80°, Arrieta quotation, plumb-bob interpretation. <https://www.borntoengineer.com/neolithic-engineers-scientific-principles>

²¹⁸García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

- **The combined reading [INFERRED, this investigation’s assessment]:** both. A Neolithic mind did not split “protective” from “meaningful.” Turning the sea-floor face inward sheltered the stone *and* clothed the chamber in ocean. The same rotation satisfies physics and symbol at once — the signature move of this entire monument, seen again in the well (water + rite) and the orientation (landmark + light) [INFERRED].

14.4 Finding

Flag 4 resolves to: **a universal, deliberate rotation of all structural stones from quarry orientation, serving demonstrable construction logic and probable protective function, with a strong symbolic resonance (marine interiors) corroborated by independent evidence from the sister monuments** [MEASURED pattern; INFERRED functions]. The universality — zero exceptions documented — is itself the loudest fact: nothing at Menga was left unchosen [MEASURED]²¹⁹.

CHAPTER 15: FLAG 5 — SOFT STONE, ON PURPOSE

FLAG STATEMENT. *The builders used soft, porous, poorly cemented calcarenites and breccias for the entire monument — including the 150-ton capstone — although harder, more durable rocks outcrop nearby and were quarried by the same communities for tools* [MEASURED]²²⁰.

15.1 The case for irrationality

On its face, the choice looks like malpractice. Soft stone is fragile in transport, weaker in compression, and — worst — **hydrophilic**: its pores wick water, adding weight and inviting granular disintegration [MEASURED]²²¹. A megalithic tomb is supposed to be forever. Forever argues for granite.

The neighboring traditions mostly agreed: the great Breton megaliths are granite; Stonehenge’s sarsens are silicified sandstone of brutal hardness. Fragile stone at 150-ton scale has no close parallel in the European record — the quarry paper notes that while soft calcarenites appear in other Iberian megaliths (e.g., Pantano de los Bermejales) and in historic buildings, “none approach Menga’s scale or employ equally enormous pieces” [DOCUMENTED]²²².

15.2 The case for genius

Now reverse the lens. What does soft stone *buy*?

1. **Workability.** Calcarenite can be cut, dressed, and faceted with stone tools. The interlocking facets, the polynomial wall, the dressed capstone bearing surfaces — the entire geometric program of Chapters 7–9 — is executable in this material and would be brutally slow in hard rock [INFERRED from MEASURED toolset and geometry].

²¹⁹Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

²²⁰Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

²²¹Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

²²²Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

2. **Quarry geometry.** The chosen beds at Cerro de la Cruz carried the favorable joint system (7.5 m vertical spacing, 55° valley-dipping partings) that made colossal blocks extractable at all [MEASURED]²²³. The soft rock was not merely available; it was *pre-engineered by tectonics*.
3. **Weight per volume.** Lower density means a given dimension of chamber can be roofed with less mass — the difference between movable and immovable at 6 × 7 meter spans [INFERRED].
4. **Managed weakness.** The builders did not ignore the fragility; they compensated for it at every level: gentle transport on prepared trackways, no lifting (the trench method), deep sockets that cradle rather than clamp, a relieving arch that reroutes stress around the weakest slab, and a waterproof tumulus that keeps the enemy — water — off the stone for six millennia [MEASURED, assembled from Chapters 6–10]^{224,225}.

The pattern is unmistakable once seen whole: **soft stone was chosen because it was the only material in which the design could be executed, and the design includes an entire subsystem — the armor mound — whose primary job is to protect the soft stone from its own weakness** [INFERRED]. Material and architecture are one decision, not two.

15.3 What the choice reveals

Flag 5 resolves to: **a deliberate, informed materials selection — a geotechnical judgment of the highest order, made before the first block was cut, and compensated by an integrated protection system.** The builders knew their plain's rocks the way a modern engineering firm knows its materials tables [INFERRED from MEASURED petrography and structure]. The anomaly is not that they chose badly. The anomaly is that they chose *correctly* in a way that requires us to credit Late Neolithic farmers with empirical materials science [MEASURED → INFERRED].

CHAPTER 16: FLAG 6 — THE DEEPEST FOUNDATIONS (AND THE ONE MISTAKE)

FLAG STATEMENT. *Orthostats and pillars are embedded in bedrock to depths of 2.75–3.2 m — about one-third of their length — far deeper than typical for gallery dolmens; the design reads as a seismic-stability system. One orthostat, O-14, shows buckling from a material flaw under load — the monument's single documented error, which the structure absorbed without failing* [MEASURED]²²⁶.

16.1 Why so deep?

The embedment exceeds any structural minimum for static loads. An orthostat socketed 50–80 cm deep stands fine under a roof it merely has to hold up. Three-meter sockets are over-engineering by a factor of three or four — unless the design load is not static [INFERRED from engineering first principles].

²²³Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

²²⁴Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

²²⁵Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

²²⁶Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

What dynamic load does the Antequera plain supply? **Earthquakes.** The Betic Cordillera is the most seismically active zone in Iberia; Antequera's province has been shaken repeatedly through recorded history. A stone chamber is an assembly of discrete blocks: its enemy is not crushing but *racking* — sideways acceleration that walks stones out of plumb and pops capstones off their seats. Deep socketing converts each orthostat from a standing slab (easy to topple) into a cantilevered pile (gripped by the earth itself). The inward lean then ensures that any racking drives stones tighter into their interlocks rather than apart [INFERRED, consistent with the research team's own reading: sockets "maximized the structure's strength and created resistance to seismic vibrations — a technique still used in modern construction"]²²⁷.

The proof of the system is not theoretical: **the monument has survived ~5,700 years in a seismic zone with its capstones still on their pillars** [MEASURED — by existence]. Menga is a fifty-seven-century duration test of its own engineering, passed.

16.2 O-14: the flaw that proves the science

Orthostat O-14 shows buckling — a failure of the stone itself, traced to unsuitable planar discontinuities in its lithology, loaded by capstone C-5 and the tumulus [MEASURED]²²⁸. Somebody, 5,700 years ago, set a stone whose hidden bedding planes ran the wrong way for the load it would carry, and the stone has been slowly failing ever since.

This investigation reads O-14 as one of the most valuable single artifacts in the monument, for three reasons [INFERRED]:

1. **It proves the loads were calculated, not just survived.** You only get a *specific* failure — one stone, at the highest-load position, for a traceable material reason — in a structure whose other 31 stones are correctly matched to their loads. Random construction fails randomly. Menga's single failure is diagnostic, and the diagnosis is materials selection, not design.
2. **It proves the builders could not fully inspect their material.** The discontinuity was invisible at the quarry face — the limit of Neolithic non-destructive testing. They were working at the edge of what their tools could verify [INFERRED].
3. **It proves the redundancy worked.** O-14 buckled; the roof did not fall. The pillar-and-interlock system absorbed a member failure — exactly what redundancy is for [MEASURED outcome].

16.3 Finding

Flag 6 resolves to: **an over-specified foundation system that functions as seismic engineering and has demonstrably performed as such for 5,700 years; plus one material-selection error, absorbed by structural redundancy** [MEASURED depths and failure; INFERRED seismic intent]. The seismic reading cannot be proven from the stones alone — no inscription says "for earthquakes" — but no competing explanation accounts for tripling the required embedment depth [INFERRED; formally OPEN as to intent, MEASURED as to performance].

²²⁷Born to Engineer — "Neolithic Engineers Used Advanced Scientific Principles..." (2024): interlock angles 77–80°, Arrieta quotation, plumb-bob interpretation. <https://www.borntoengineer.com/neolithic-engineers-scientific-principles>

²²⁸Lozano Rodríguez, J.A., García Sanjuán, L., et al. — "Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain)," *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

CHAPTER 17: FLAG 7 — THE MISSING LADDER

FLAG STATEMENT. *Menga appears, at 3800–3600 BCE, as the oldest of the great Iberian dolmens, with engineering (interlocking facets, deep sockets, relieving arch, hydraulic integration) that has no identified local precedents of comparable scale or sophistication — and no regional chain of smaller “practice” monuments demonstrating a learning curve [MEASURED/DOCUMENTED]²²⁹²³⁰.*

17.1 The shape of the problem

The dating team titled their discussion section “Birth of a Giant” — and the word *birth* is precise. Before Menga on the Antequera plain: cairns over single graves (Arroyo Saladillo, 4037–3805 BCE) [MEASURED]²³¹. After Menga: Viera, then a millennium later El Romeral. At Menga itself: a fully formed colossal engineering system, first of its kind, oldest of the great Iberian dolmens [MEASURED]²³².

Innovation normally leaves a sediment of trials: prototypes, partial successes, collapsed experiments. Around Antequera, the ladder between “stone cairn” and “1,140-ton interlocked chamber with internal well” is missing its rungs [DOCUMENTED — absence noted]²³³. The *Science Advances* authors state it flatly: Menga was “designed as a completely original engineering project, for which we know of no precedents in Iberia” [DOCUMENTED]²³⁴.

17.2 Four explanations, weighed

(a) The ladder existed but is lost or unfound. Absence of evidence is chronic in prehistory; mounds get plowed, quarried, built over. A 5-km radius around Antequera is among the most archaeologically scrutinized landscapes in Iberia, yet Piedras Blancas — a whole megalithic tomb — hid under the giant’s chest until 2020 [DOCUMENTED]²³⁵. Verdict: possible; unfalsifiable; keeps the flag [OPEN] but does not retire it.

(b) Knowledge arrived by mobility. The dating team itself raises this: the early fourth millennium saw supra-regional interaction across Europe — variscite (a green gemstone) from Huelva, 200 km west, reached Carnac in Brittany by c. 4500 BCE [MEASURED/DOCUMENTED]²³⁶. Atlantic France had monumental megalithic traditions (mounds, coffres, menhirs) from c. 4700 BCE, centuries before Iberia’s

²²⁹University of Alcalá repository record for the *Science Advances* paper, including the official abstract (“no precedents in Iberia”). <https://ebuah.uah.es/dspace/handle/10017/63692?show=full>

²³⁰García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²³¹García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²³²García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²³³University of Alcalá repository record for the *Science Advances* paper, including the official abstract (“no precedents in Iberia”). <https://ebuah.uah.es/dspace/handle/10017/63692?show=full>

²³⁴University of Alcalá repository record for the *Science Advances* paper, including the official abstract (“no precedents in Iberia”). <https://ebuah.uah.es/dspace/handle/10017/63692?show=full>

²³⁵García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña’s geology and the 45° alignment’s cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

²³⁶García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

first proto-megaliths [MEASURED]²³⁷. If one seeks a “foreign inspiration,” the literature says, Atlantic France is the obvious reference [INFERRED by the authors]²³⁸. But note what the French tradition does *not* contain: polygonal interlocking chambers on pillars, 150-ton relieving arches, chamber wells. Travelers could carry the *idea* of stone monumentality; they could not carry Menga’s engineering, which has no French prototype either [INFERRED]. Verdict: explains the impulse, not the technique.

(c) Local invention under pressure. The El Toro cave collapse (c. 4000 BCE) destroyed the region’s ancestral shelter within living memory of the project [MEASURED]²³⁹. A society that loses its stone-roofed holy place to gravity may develop, in a generation or two, an obsessive competency in stone roofs [INFERRED — this investigation flags the narrative elegance and the evidential softness equally]. Verdict: unfalsifiable as psychology, but chronologically real as context [OPEN].

(d) Convergence of a knowledge commons. Antequera sat at the crossroads (Chapter 1), with flint, salt, ochre, marble, and hard toolstone drawing exchange; the plain supported dense farming settlement [MEASURED/DOCUMENTED]²⁴⁰. Crossroads concentrate not just goods but *craftsmen* — knappers, ropemakers, carpenters, stone-workers — and crossroads are where craft traditions hybridize. On this reading, the “missing ladder” is not architectural but social: the learning happened in perishable media (timber trackways, rope rigging, wooden templates) across a few decades of concentrated collaboration, and Menga is the first project big enough to make the knowledge visible in stone [INFERRED — this investigation’s own reading, offered as such]. Verdict: consistent with all evidence; unproven.

17.3 Finding

Flag 7 stays **open**. The measured facts are these: the monument is the oldest of its class [MEASURED]; its engineering is original in the Iberian record [DOCUMENTED]; its sophistication exceeds every local predecessor by orders of magnitude [MEASURED]. Any honest model must explain a discontinuity. This investigation records the discontinuity without filling it — and notes that the discontinuity itself is the finding: **whatever happened at Antequera around 3800 BCE, it was not gradual** [MEASURED].

CHAPTER 18: FLAG 8 — THE THOUSAND-YEAR STRAIGHT LINE

FLAG STATEMENT. *The three great monuments of the complex — Menga, Viera, and El Romeral — together with La Peña de los Enamorados, lie on or along a single visual axis spanning roughly a millennium of construction* [DOCUMENTED/MEASURED]^{241 242}.

²³⁷García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²³⁸García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²³⁹García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²⁴⁰Museos de Andalucía — official site brochure (English): research history from 1530 onward, El Toro sequence, settlement catalog, Hoskin’s 2002 confirmations. <https://www.museosdeandalucia.es/documents/1971642/0/Folieto+en+ingl%C3%A9s/>

²⁴¹Antequera Tourism — official monument dossier: the two-line orientation system (Menga→La Peña; Viera→equinox; El Romeral on the axis, → El Torcal). <https://turismo.antequera.es/wp-content/uploads/2011/06/DOLMENES-INGLES.pdf>

²⁴²Museos de Andalucía — official site brochure (English): research history from 1530 onward, El Toro sequence, settlement catalog, Hoskin’s 2002 confirmations. <https://www.museosdeandalucia.es/documents/1971642/0/Folieto+en+ingl%C3%A9s/>

18.1 The geometry of the complex

Stand at the El Romeral tholos (Copper Age, ~3rd millennium BCE) and look northeast: you are looking along the trajectory Viera → Menga → La Peña [DOCUMENTED]²⁴³²⁴⁴. The visual axis was described in print by Manuel Gómez-Moreno in 1905 and photographed by Wilfrid Hemp in 1934; Hoskin confirmed the orientations instrumentally in 2002 [DOCUMENTED]²⁴⁵.

Each monument's own facing tells its chapter of the story [MEASURED]²⁴⁶²⁴⁷:

- **Menga:** axis 45°, to La Peña's chin. Summer-solstice light effect inside.
- **Viera:** corridor east, equinoctial sunrise. The obedient tomb, obeying Hoskin's law — built in Menga's shadow.
- **El Romeral:** corridor 199° southwest, to the Camorro de las Siete Mesas in El Torcal; winter-solstice light reaches the inner chamber. One of the few west-facing monuments in Iberia.

Together the complex keys four calendrical anchor points — summer solstice (Menga), equinoxes (Viera), winter solstice (El Romeral) — and two terrestrial giants (La Peña, El Torcal) [MEASURED]²⁴⁸. UNESCO's framing: an astronomical annual cycle worked into the landscape, a combination found nowhere else on the World Heritage List [DOCUMENTED]²⁴⁹.

18.2 What a straight line across a millennium means

Menga was ancient — a thousand years old, already older to them than El Romeral is to us — when El Romeral's builders placed their tholos *on its sight-line* [MEASURED chronology; INFERRED intent]. That means the axis was not a construction convenience. It was a **memory**: a transmitted, surveyed, curated line across the plain, honored by people separated from Menga's builders by as much time as separates us from William the Conqueror [INFERRED].

The straight line is therefore evidence of something no single monument could show: **continuity of knowledge** — that positions, orientations, and meanings were curated across fifty generations without writing. Whether by priesthood, lineage, festival, or survey tradition, we cannot say [OPEN]. That it happened is written in the ground [MEASURED].

18.3 Finding

Flag 8 resolves to: **a deliberately maintained inter-monument axis integrating two natural monuments and four astronomical events, executed in three architectures across ~1,000 years** [MEASURED/DOCUMENTED]. It is the strongest evidence at Antequera for institutional memory — for a *school*, in the broadest sense, that outlived its founders by a millennium [INFERRED].

²⁴³Antequera Tourism — official monument dossier: the two-line orientation system (Menga→La Peña; Viera→equinox; El Romeral on the axis, → El Torcal). <https://turismo.antequera.es/wp-content/uploads/2011/06/DOLMENES-INGLES.pdf>

²⁴⁴Museos de Andalucía — official site brochure (English): research history from 1530 onward, El Toro sequence, settlement catalog, Hoskin's 2002 confirmations. <https://www.museosdeandalucia.es/documents/1971642/0/Folieto+en+ingl%C3%A9s/>

²⁴⁵Antequera Tourism — official monument dossier: the two-line orientation system (Menga→La Peña; Viera→equinox; El Romeral on the axis, → El Torcal). <https://turismo.antequera.es/wp-content/uploads/2011/06/DOLMENES-INGLES.pdf>

²⁴⁶Antequera Tourism — official monument dossier: the two-line orientation system (Menga→La Peña; Viera→equinox; El Romeral on the axis, → El Torcal). <https://turismo.antequera.es/wp-content/uploads/2011/06/DOLMENES-INGLES.pdf>

²⁴⁷Antequera World Heritage — Tholos de El Romeral: corbelled construction, 199° orientation to Camorro de las Siete Mesas, winter-solstice light. <https://www.antequerapatrimoniomundial.com/en/antequera-dolmens/tholos-el-romeral/>

²⁴⁸Go-Torcal-Antequera — UNESCO site guide: the four-event astronomical cycle; Viera equinox; El Romeral 199°/winter solstice. <https://www.go-torcal-antequera.com/en/the-el-torcal/unesco-world-heritage-antequera-dolmens-site/>

²⁴⁹Go-Torcal-Antequera — UNESCO site guide: the four-event astronomical cycle; Viera equinox; El Romeral 199°/winter solstice. <https://www.go-torcal-antequera.com/en/the-el-torcal/unesco-world-heritage-antequera-dolmens-site/>

CHAPTER 19: FLAG 9 — TOMB, TEMPLE, OR MACHINE?

FLAG STATEMENT. *Menga is conventionally classified as a collective burial chamber, and 19th-century accounts describe the skeletons of several hundred individuals found inside — yet its engineering (well, light effects, landscape targeting, seismic redundancy) exceeds any functional requirement of a tomb [REPORTED/MEASURED]²⁵⁰²⁵¹.*

19.1 The burial evidence, honestly weighed

The claim that several hundred skeletons were found inside when the monument was examined in the 19th century is repeated across the modern literature and reference works [REPORTED]²⁵²²⁵³. But the pedigree matters: the chamber had been open and visited for centuries — possibly since Roman times [INFERRED]²⁵⁴ — before any controlled excavation. Mitjana’s own 1847 report describes digging in the chamber center where remains “were believed to lie,” and finding the shaft area disturbed or empty [DOCUMENTED]²⁵⁵. The “hundreds of skeletons” figure descends from early accounts of the chamber’s contents at first opening, not from a modern osteological count; no curated Neolithic assemblage of that size from Menga exists in a museum today [REPORTED; the original collection’s fate is poorly DOCUMENTED].

What modern excavation *has* established: use-deposits from the Copper Age, Bronze Age, Iron Age, Roman period, Middle Ages, and modern era [MEASURED/DOCUMENTED]²⁵⁶; Roman-era tombs around the dolmen [MEASURED]²⁵⁷; two medieval burials (8th–11th c.) in the atrium, aligned with the monument’s axis [MEASURED]²⁵⁸. Funerary use is certain; the scale and date of the *primary* funerary deposit is largely lost to the monument’s long accessibility [OPEN].

19.2 The excess problem

Now measure the gap between function and construction. A collective tomb requires: a sealed-ish chamber, some floor area, an entrance. Menga supplies, in addition [MEASURED]: a well to the aquifer; a solstice light instrument; a mountain-targeted axis; seismic over-engineering; a relieving arch; a water-proof armored mound; a place on a thousand-year survey line. None of these bury anyone better.

The engineering excess is the flag. It means the building’s *brief* was bigger than burial [INFERRED]. The most defensible expansion of the brief, on the evidence: a monument that anchored a community’s

²⁵⁰Wikipedia — “Dolmen of Menga” (reference summary; the “several hundred skeletons” tradition is [REPORTED] through this and the following). https://en.wikipedia.org/wiki/Dolmen_of_Menga

²⁵¹Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

²⁵²Wikipedia — “Dolmen of Menga” (reference summary; the “several hundred skeletons” tradition is [REPORTED] through this and the following). https://en.wikipedia.org/wiki/Dolmen_of_Menga

²⁵³@archeohistories — monument summary thread (X), the original prompt’s source for the popular framing; superseded by primary literature throughout this book. <https://x.com/archeohistories/status/1708781792252338569>

²⁵⁴Hobthrush — “Dolmen de Menga”: the well’s early-1700s suppression, ~50 t of fill, animal carcasses, carved crosses, and the pre-suppression clearing hypothesis (tagged [REPORTED]/[INFERRED] in the main text). <https://hobthrush.co.uk/dolmen-de-menga-antequera-spain/>

²⁵⁵Andalucia.com — “Dolmens of Antequera”: Mitjana’s 1842–47 dig and shaft note; Tenison 1852/1853 warning; Carrión’s 2005 well excavation; 19th-century visitor list. <https://www.andalucia.com/antequera/dolmens-de-menga.htm>

²⁵⁶García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²⁵⁷García Sanjuán, L., et al. — author manuscript of the multimethod dating paper (University of Southampton ePrints), including mound dimensions and the catalog of historical radiocarbon determinations. https://eprints.soton.ac.uk/470780/1/Garcia_Sanjuana_et_al_2022_A_multimethod_approach_to_the_genesis_of_menga_a_world_heritage_megalith.pdf

²⁵⁸Medievalists.net — “Two Medieval Men Found Buried in Prehistoric Site” (2026), reporting Silva et al., *Journal of Archaeological Science: Reports* 69 (2026): the axis-aligned 8th–11th c. burials. <https://www.medievalists.net/2026/03/two-medieval-men-found-buried-in-prehistoric-site/>

claims — to the mountain (the axis), to the calendar (the light), and to the plain’s scarcest resource (the water) — with the dead installed as permanent witnesses and title-holders [INFERRED; the primacy among these functions is OPEN].

19.3 Finding

Flag 9 resolves to: **funerary use is [MEASURED/REPORTED] and primary classification as a tomb is fair — but the structure is a multi-function instrument of which burial is one documented layer** [INFERRED from MEASURED feature set]. The “several hundred skeletons” figure is [REPORTED], not [MEASURED], and should be quoted with that tag wherever it appears.

CHAPTER 20: FLAG 10 — THE SCALE EXTREMES

FLAG STATEMENT. *Menga holds the Iberian record for the heaviest stone moved in the megalithic phenomenon (C-5, ~150 t), totals ~1,140 t across 32 stones, and is the largest stone monument built in Europe in its time — with its largest elements exceeding anything at Stonehenge by a factor of four* [MEASURED]²⁵⁹²⁶⁰²⁶¹.

20.1 The record table

Metric	Menga	Context	Tag
Heaviest element	C-5, ~149.6 ± 9.2 t	Largest ever in an Iberian megalith; 2nd in Europe after Grand Menhir Brisé (~335 t) —but the Grand Menhir is a standing stone, not a composite-structure component	[MEASURED] ²⁶²
Total structural mass	~1,140 t (32 stones)	Unmatched in Iberian megalithism	[MEASURED] ²⁶³
Chamber length	~24.9–27.5 m (with atrium)	Among Europe’ s largest chambers	[MEASURED] ²⁶⁴²⁶⁵

²⁵⁹Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

²⁶⁰Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

²⁶¹@archohistories — monument summary thread (X), the original prompt’s source for the popular framing; superseded by primary literature throughout this book. <https://x.com/archohistories/status/1708781792252338569>

²⁶²Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

²⁶³Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

²⁶⁴Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

²⁶⁵Wikipedia — “Dolmen of Menga” (reference summary; the “several hundred skeletons” tradition is [REPORTED] through this and the following). https://en.wikipedia.org/wiki/Dolmen_of_Menga

Table 1 – continued

Metric	Menga	Context	Tag
Interior height	2.5 m → 3.45 m	Rising floor-to-roof gradient by design	[MEASURED] ²⁶⁶
Tumulus	48.3 × 44 m, ~2,730 m ²	Armored composite, not soil heap	[MEASURED] ²⁶⁷
Well	19.4 m deep, 1.5 m diameter	Unique in megalithic Europe	[MEASURED] ²⁶⁸
Date	c. 3800–3600 BCE	Oldest of the great Iberian dolmens	[MEASURED] ²⁶⁹

For calibration: Stonehenge’s largest sarsens weigh ~25–30 t; its bluestones 2–5 t [DOCUMENTED]. The heaviest single lift at Menga is five Stonehenge sarsens at once — in softer stone, a thousand years earlier.

20.2 Why scale is itself a flag

Scale is not an anomaly because it is big; it is an anomaly because of *when* and *where first*. If Menga were the culmination of a millennium of growing Iberian dolmens, its size would be a trend’s endpoint. Instead it is the **first** of the great dolmens [MEASURED]²⁷⁰. The record-setter is the pioneer. In the history of technology, that is the signature of a discontinuity — and it converts Flag 10 from a superlative into corroboration of Flag 7’s missing ladder [INFERRED].

20.3 Finding

Flag 10 resolves to: **extremes that are [MEASURED] and uncontested, whose anomalous character lies in their earliness and primacy rather than their size alone**. Nothing at this scale would be attempted again in Iberia for a very long time; in several respects (composite-stone mass, chamber hydraulics), never again in the megalithic world [DOCUMENTED/INFERRED]²⁷¹.

End of Part III. The Flag Register in Appendix A consolidates all ten flags with their evidence tiers. Part IV follows the monument through its five and a half millennia of afterlife.

²⁶⁶Lozano Rodríguez, J.A., García Sanjuán, L., et al. —“Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

²⁶⁷García Sanjuán, L., et al. —author manuscript of the multimethod dating paper (University of Southampton ePrints), including mound dimensions and the catalog of historical radiocarbon determinations. https://eprints.soton.ac.uk/470780/1/Garcia_Sanjuan_et_al_2022_A_multimethod_approach_to_the_genesis_of_menga_a_world_heritage_megalith.pdf

²⁶⁸García Sanjuán, L., Balsera Nieto, V., et al. —“A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’ s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²⁶⁹García Sanjuán, L., Balsera Nieto, V., et al. —“A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’ s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²⁷⁰García Sanjuán, L., Balsera Nieto, V., et al. —“A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’ s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²⁷¹University of Alcalá repository record for the *Science Advances* paper, including the official abstract (“no precedents in Iberia”). <https://ebuah.uah.es/dspace/handle/10017/63692?show=all>

PART IV — THE LONG AFTERLIFE

A monument is not finished when the builders leave. Menga has been in continuous use, memory, and dispute for 5,700 years. That record is itself evidence.

CHAPTER 21: THE NEVER-FORGOTTEN MONUMENT

21.1 A unique biographical fact

Most megaliths were forgotten at some point — sealed, plowed under, rediscovered by antiquarians. Menga, the dating team notes, “appears to have been in constant use since its original construction, never falling into oblivion” [DOCUMENTED]²⁷². The stratigraphy reads like a visitor’s book signed by every era: Copper Age, Bronze Age, Iron Age, antiquity, Middle Ages, modern history [MEASURED deposits]²⁷³.

21.2 The afterlife in layers

Copper Age and Bronze Age (c. 3500–1200 BCE): The plain’s monument-building continued — El Romeral was raised on Menga’s sight-line a millennium after Menga (Chapter 18). Activity deposits at the dolmen itself, and OSL ages from atrium ceramics, attest repeated visits [MEASURED]²⁷⁴. When the great Argaric Bronze Age culture rose in southeastern Iberia, Antequera’s monuments were already a thousand years old — inherited giants in a landscape their authors had become “a powerful past” to live on the wake of [DOCUMENTED]¹⁸, referencing García Sanjuán & Mora Molina 2022].

Roman period: Eight radiocarbon determinations from tombs and structures immediately around the dolmen document significant activity in antiquity — the Romans buried their own dead beside the ancient chamber, a classic appropriation of an ancestral place [MEASURED]²⁷⁵²⁷⁶.

Early medieval period: The 2005 excavation found two men buried in simple pits in the dolmen’s atrium, dated to between the 8th and 11th centuries, about 190 years apart — repeated, episodic use rather than a cemetery [MEASURED]²⁷⁷. A 2026 archaeogenetic study (Universities of Seville, Huddersfield, the Francis Crick Institute, Harvard) adds the remarkable details: both were older men (45+), buried face down, heads to the southwest — and **both burials were carefully aligned with the**

²⁷²García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²⁷³García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²⁷⁴García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²⁷⁵García Sanjuán, L., et al. — author manuscript of the multimethod dating paper (University of Southampton ePrints), including mound dimensions and the catalog of historical radiocarbon determinations. https://eprints.soton.ac.uk/470780/1/Garcia_Sanjuan_et_al_2022_A_multimethod_approach_to_the_genesis_of_menga_a_world_heritage_megalith.pdf

²⁷⁶García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

²⁷⁷Medievalists.net — “Two Medieval Men Found Buried in Prehistoric Site” (2026), reporting Silva et al., *Journal of Archaeological Science: Reports* 69 (2026): the axis-aligned 8th–11th c. burials. <https://www.medievalists.net/2026/03/two-medieval-men-found-buried-in-prehistoric-site/>

dolmen's axis of symmetry [MEASURED]²⁷⁸. Genetic ancestry pointed toward contacts with North Africa and the Levant consistent with post-8th-century Iberia [MEASURED]²⁷⁹. Four thousand years after construction, people burying their dead here still knew — or still obeyed — the line [INFERRED: knowledge or conformity to the axis; the mechanism is OPEN].

Late medieval and early modern: The “Cueva de Menga” enters written records in a Málaga bishop’s document of 1530 — *antrum de Menga vulgariter nuncupatum*, “the cave commonly called Menga” [DOCUMENTED]²⁸⁰. By the late 1500s the cave is described as a place of pagan worship and “demonic rituals,” its builders assumed supernatural [REPORTED]²⁸¹. The well’s biography (Chapter 13.4) belongs to this era: healing-water visits, water jars, carved crosses, and finally the early-1700s suppression with 50 tons of fill and animal carcasses [MEASURED/DOCUMENTED]²⁸²²⁸³.

21.3 What continuity proves

An object that remains meaningful across 5,700 years — through the fall of the cultures that built it, through Rome, through Islam, through Christian reconquest, through the Enlightenment — is not merely durable. It is *legible*: every era could read something into it (tomb, temple, fortress of giants, healing place, wonder) [INFERRED]. The continuity is the monument’s second engineering triumph: it was built not just to last, but to keep making sense [INFERRED].

CHAPTER 22: THE AGE OF LEGENDS

Legends are evidence — not of what happened in the Neolithic, but of what the monument did to the minds that inherited it. This investigation reads them as data about perception, tagged [REPORTED].

22.1 Menga: the woman in the stone

The name itself is a fossil of incomprehension. Linguistically, *Menga* is a folk shortening of **Dominga** — a common medieval woman’s name (from Domingo, “Sunday,” literally “the Lord’s day”). The form appears as a nickname in Castilian texts of the 13th–16th centuries; the cave’s first written attestation is the 1530 bishop’s document [DOCUMENTED]²⁸⁴. The monument is not named for a prehistoric anything. It is named for a medieval story invented to explain it.

The story, written down in its full form by the Antequera chronicler Trinidad de Rojas in the newspaper

²⁷⁸Medievalists.net — “Two Medieval Men Found Buried in Prehistoric Site” (2026), reporting Silva et al., *Journal of Archaeological Science: Reports* 69 (2026): the axis-aligned 8th–11th c. burials. <https://www.medievalists.net/2026/03/two-medieval-men-found-buried-in-prehistoric-site/>

²⁷⁹Archaeology Mag — “Two medieval burials in Spain’s Menga dolmen reveal its lasting sacred role” (2025). <https://archaeologymag.com/2025/12/medieval-burials-in-spain-s-menga-dolmen/>

²⁸⁰Go-Torcal-Antequera — “Where a Medieval Legend Explains a 19-Metre-Deep Shaft”: the Menga/Dominga etymology, the 1530 document, Rojas 1874, the leper and sorceress variants. <https://www.go-torcal-antequera.com/en/magazine/legend-woman-menga-dolmen/>

²⁸¹Hobthrush — “Dolmen de Menga”: the well’s early-1700s suppression, ~50 t of fill, animal carcasses, carved crosses, and the pre-suppression clearing hypothesis (tagged [REPORTED]/[INFERRED] in the main text). <https://hobthrush.co.uk/dolmen-de-menga-antequera-spain/>

²⁸²Hobthrush — “Dolmen de Menga”: the well’s early-1700s suppression, ~50 t of fill, animal carcasses, carved crosses, and the pre-suppression clearing hypothesis (tagged [REPORTED]/[INFERRED] in the main text). <https://hobthrush.co.uk/dolmen-de-menga-antequera-spain/>

²⁸³García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well’s excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

²⁸⁴Go-Torcal-Antequera — “Where a Medieval Legend Explains a 19-Metre-Deep Shaft”: the Menga/Dominga etymology, the 1530 document, Rojas 1874, the leper and sorceress variants. <https://www.go-torcal-antequera.com/en/magazine/legend-woman-menga-dolmen/>

Genil on 14 February 1874, runs thus [REPORTED]²⁸⁵: in the 16th century, a noble widow named Margarita — some versions say Dominga — is struck by leprosy. Banished from society, she takes refuge in the mysterious half-ruined stone cave at the edge of town. Inside, she finds a deep well whose waters, the tale insists, have healing power. She bathes her wounds, drinks, and lives on in the cave — some versions say healed, all versions say until her death. Afterward the cave bears her name: Cueva de Dominga → Cueva de Menga.

A parallel version makes Menga a sorceress who built the dolmen by supernatural power, in rivalry with **Mari Macho**, a female figure attached to the Cerro de Marimacho hill nearby — two mysterious women explaining two mysterious hills [REPORTED]²⁸⁶.

Read as data: the legend encodes three true things about the site — the well, the water’s special reputation, and the medieval conviction that no human hand could have raised such stones [REPORTED → INFERRED reading]. The pattern is classic: faced with achievement beyond its own frame of reference, the medieval mind supplied a woman, an illness, and a miracle.

22.2 The lovers of the Peña

The mountain collected its own legend, later and literary. In the best-known version — set in the early 15th century, in the last decades of Moorish Antequera — a Christian soldier, **Tello**, imprisoned in the Alcazaba, and **Tagzona (Tazgona)**, daughter of a Moorish noble of Archidona, fall in forbidden love. Fleeing between two armies, they climb the table mountain and, cornered at the summit at sunrise, leap hand in hand from the southern cliff [REPORTED]²⁸⁷. Robert Southey gave the tale literary form in *Laila and Manuel*, recasting the lovers as a Muslim girl and a Christian slave [DOCUMENTED]²⁸⁸.

The legend is medieval romance layered onto something much older — and the proof of the older layer is that the mountain’s *meaning* (an anthropomorphic presence worth orienting sacred architecture toward) is 5,800 years old, while the lovers are fifteen centuries young [MEASURED antiquity vs. REPORTED legend]. The story changed; the sleeping figure did not.

A coda that certifies the mountain’s fame: on 29 October 1492, Christopher Columbus, describing the Cuban coast, compared its mountains to “la Peña de las enamoradas” — the reference recorded by his chronicler Bartolomé de las Casas [DOCUMENTED]²⁸⁹. The sleeping face of Antequera sailed to the New World as a simile.

22.3 What the legends tell the investigation

Three legend-strata, three responses to the same stimulus: the monument as (1) healer’s cave, (2) sorceress’s work, (3) lover’s tomb-mountain. All three agree on the premise — **this was not made by ordinary people in the ordinary way** [REPORTED → INFERRED]. The folklore is wrong about the mechanism and right about the anomaly. In its own uncalibrated way, tradition detected exactly

²⁸⁵Go-Torcal-Antequera — “Where a Medieval Legend Explains a 19-Metre-Deep Shaft”: the Menga/Dominga etymology, the 1530 document, Rojas 1874, the leper and sorceress variants. <https://www.go-torcal-antequera.com/en/magazine/legend-woman-menga-dolmen/>

²⁸⁶Go-Torcal-Antequera — “Where a Medieval Legend Explains a 19-Metre-Deep Shaft”: the Menga/Dominga etymology, the 1530 document, Rojas 1874, the leper and sorceress variants. <https://www.go-torcal-antequera.com/en/magazine/legend-woman-menga-dolmen/>

²⁸⁷Go-Torcal-Antequera — “The Legend of the Lovers of the Peña”: the Tello and Tagzona narrative and its variants. <https://www.go-torcal-antequera.com/en/magazine/legend-lovers-pena-de-los-enamorados/>

²⁸⁸Kiddle — “Peña de los Enamorados”: legend variants, Southey’s *Laila and Manuel*, Columbus’s 1492 comparison. https://kids.kiddle.co/Pe%C3%B1a_de_los_Enamorados

²⁸⁹Kiddle — “Peña de los Enamorados”: legend variants, Southey’s *Laila and Manuel*, Columbus’s 1492 comparison. https://kids.kiddle.co/Pe%C3%B1a_de_los_Enamorados

the flags this book has measured: the too-big stones, the too-deep well, the too-strange mountain bond [INFERRED].

CHAPTER 23: THE REDISCOVERY — AND THE MODERN INSTRUMENTS

23.1 Mitjana and the druids

The first recorded exploration was the work of **Rafael Mitjana y Ardison**, municipal architect of Málaga, who excavated from 1842 to 1847 and published the *Memoria sobre el templo druida hallado en las cercanías de la ciudad de Antequera* (“Report on the druid temple found near Antequera”) in 1847 — one of the first scientific monographs on any megalithic monument anywhere [DOCUMENTED]²⁹⁰²⁹¹.

Mitjana did what investigators do: he dug where the bodies were supposed to be. In the chamber’s center, under a large stone, he opened the shaft — and found nothing, the well having been backfilled a century and a half earlier [DOCUMENTED]²⁹²²⁹³. He also drove a crude perforation through the backstone and the mound behind it, a wound whose geometry — documented by a chain of illustrators from De Rojas (1861) to Paris (1921) — would 180 years later serve the *Science Advances* team as a fixed reference for reconstructing the bedrock topography [DOCUMENTED]²⁹⁴. Even the damage became data.

Mitjana’s “druid” attribution was the standard error of his century — every megalith was Celtic then — but his monograph made Antequera a fixed point on the European antiquarian map. The visitors and describers followed: Lady Louisa Tenison (1851/1853), who first saw and wrote the Menga–La Peña visual bond [DOCUMENTED]²⁹⁵; the Baron de Bonstetten (1865); Lord Talbot of Malahide (1870); James Fergusson (1872); Émile Cartailhac (1886); Henri and Louis Siret (1887). Jean d’Estienne’s 1878 verdict entered the permanent literature: “the most beautiful and perfect of all known dolmens” [DOCUMENTED]²⁹⁶.

23.2 The Viera brothers and the twentieth century

Between 1903 and 1905, two municipal gardeners, **Antonio and José Viera Fuentes**, traced references to a second, inaccessible “cave” and excavated it at their own expense — the passage grave that now bears their name — and then located a third monument at Cerrillo Blanco: the tholos of El Romeral [DOCUMENTED]²⁹⁷. The complete complex as we know it is thus partly the work of two self-funded amateurs. The century’s scholarship — Velázquez Bosco and Gómez-Moreno (1905), Obermaier, Paris,

²⁹⁰Andalucia.com — “Dolmens of Antequera”: Mitjana’s 1842–47 dig and shaft note; Tenison 1852/1853 warning; Carrión’s 2005 well excavation; 19th-century visitor list. <https://www.andalucia.com/antequera/dolmens-de-menga.htm>

²⁹¹Museos de Andalucía — official site brochure (English): research history from 1530 onward, El Toro sequence, settlement catalog, Hoskin’s 2002 confirmations. <https://www.museosdeandalucia.es/documents/1971642/0/Folieto+en+ingl%C3%A9s/>

²⁹²Andalucia.com — “Dolmens of Antequera”: Mitjana’s 1842–47 dig and shaft note; Tenison 1852/1853 warning; Carrión’s 2005 well excavation; 19th-century visitor list. <https://www.andalucia.com/antequera/dolmens-de-menga.htm>

²⁹³Hobthrush — “Dolmen de Menga”: the well’s early-1700s suppression, ~50 t of fill, animal carcasses, carved crosses, and the pre-suppression clearing hypothesis (tagged [REPORTED]/[INFERRED] in the main text). <https://hobthrush.co.uk/dolmen-de-menga-antequera-spain/>

²⁹⁴Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

²⁹⁵Museos de Andalucía — official site brochure (English): research history from 1530 onward, El Toro sequence, settlement catalog, Hoskin’s 2002 confirmations. <https://www.museosdeandalucia.es/documents/1971642/0/Folieto+en+ingl%C3%A9s/>

²⁹⁶Andalucia.com — “Dolmens of Antequera”: Mitjana’s 1842–47 dig and shaft note; Tenison 1852/1853 warning; Carrión’s 2005 well excavation; 19th-century visitor list. <https://www.andalucia.com/antequera/dolmens-de-menga.htm>

²⁹⁷Andalucia.com — “Dolmens of Antequera”: Mitjana’s 1842–47 dig and shaft note; Tenison 1852/1853 warning; Carrión’s 2005 well excavation; 19th-century visitor list. <https://www.andalucia.com/antequera/dolmens-de-menga.htm>

de Mortillet, Mergelina, Hemp, Giménez Reyna, and the monumental corpus of Georg and Vera Leisner — progressively mapped, planned, and contextualized the three monuments [DOCUMENTED]²⁹⁸.

23.3 The instrument age

The modern phase of the investigation — the one this book rests on — is datable and instrumented:

- **1980s–1991:** University of Málaga excavations in and around the chamber, including the mound area over C-5 — the photographs that revealed the capstone’s full convex morphology [DOCUMENTED]²⁹⁹³⁰⁰.
- **2002:** Hoskin’s instrumental confirmation of the three orientations [DOCUMENTED]³⁰¹.
- **2005:** TDTEC’s full laser-scan of Menga — the first 3D model of an Iberian megalith — the dataset behind the lean angles, the wall curves, the placement sequence [MEASURED]³⁰²³⁰³.
- **2005–2006:** University of Granada excavations (Trench 1 in the mound; the atrium; and the full excavation of the well shaft to 19.4 m, with David García González descending the bore) [DOCUMENTED]³⁰⁴³⁰⁵.
- **2016, 15 July:** UNESCO inscribes the Antequera Dolmens Site — three monuments *and two natural monuments*, a first — at the 40th session of the World Heritage Committee [DOCUMENTED]³⁰⁶.
- **2023:** the multimethod dating paper (*Quaternary Research*) fixes construction at 3800–3600 BCE; the *Scientific Reports* quarry paper fixes provenance at Cerro de la Cruz [MEASURED]³⁰⁷³⁰⁸.
- **2024:** *Science Advances* publishes the geoarchaeological synthesis — the inside-out build, the box foundations, the relieving arch [MEASURED/INFERRED]³⁰⁹.
- **2020–2026:** Piedras Blancas excavated and published (*Antiquity*); the medieval burials’ genomes

²⁹⁸Museos de Andalucía — official site brochure (English): research history from 1530 onward, El Toro sequence, settlement catalog, Hoskin’s 2002 confirmations. <https://www.museosdeandalucia.es/documents/1971642/0/Folieto+en+ingl%C3%A9s/>

²⁹⁹Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

³⁰⁰Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

³⁰¹Antequera Tourism — official monument dossier: the two-line orientation system (Menga→La Peña; Viera→equinox; El Romeral on the axis, → El Torcal). <https://turismo.antequera.es/wp-content/uploads/2011/06/DOLMENES-INGLES.pdf>

³⁰²NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

³⁰³Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

³⁰⁴García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

³⁰⁵García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well’s excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

³⁰⁶UNESCO World Heritage Centre — “Antequera Dolmens Site,” inscription 1501 (2016), criteria (iii) and (iv). <https://whc.unesco.org/en/list/1501/>

³⁰⁷García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3’s socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

³⁰⁸Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

³⁰⁹Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

published (*JAS: Reports*) [DOCUMENTED/MEASURED]³¹⁰³¹¹.

The pattern to notice: **the monument has yielded its secrets in proportion to the instruments aimed at it** — theodolite, laser scanner, petrographic microscope, accelerator mass spectrometer, and now genome sequencer [INFERRED]. There is no reason to believe the yield is finished. That observation frames Part V.

PART V — THE ASSESSMENT

CHAPTER 24: WHAT THE EVIDENCE COMPELS

This investigation set out with no adopted hypotheses and a single method: follow the measured anomalies. Here is where they led. Each conclusion is stated at its proven confidence level.

Conclusion 1 — Dating: [MEASURED], settled. Menga was built between c. 3800 and 3600 BCE, by Bayesian-modeled radiocarbon on mound-stratigraphy samples, corroborated independently by OSL ages from Pillar 3's socket infill ³¹². It is the oldest of the great Iberian dolmens. Claims of later construction contradict the radiometric record and are excluded.

Conclusion 2 — Builders: [MEASURED] + [INFERRED], settled at the level of population, open at the level of organization. The builders were the Late Neolithic communities of the Antequera plain — local, numerous, organized, and in possession of a mature empirical engineering tradition. No evidence supports exotic or non-local builders; no evidence requires them. The social form of the organization (chiefdom, confederation, ritual sodality) is [OPEN].

Conclusion 3 — Engineering: [MEASURED], the strongest finding of this book. Menga is a designed system: provenance-matched soft stone; downhill one-kilometer logistics; trench-built chamber eliminating vertical lifts; inside-out counterweight erection; 84–85° interlocked walls; 3-meter box foundations; redundant pillar-supported roof; the earliest known relieving arch; a waterproof armored tumulus. Every element is documented; the system's seismic and hydraulic performance is demonstrated by 5,700 years of survival. The design integrates solutions to transport, statics, dynamics, hydrology, and optics in a single coherent project with no known Iberian precedent ³¹³³¹⁴³¹⁵. Whatever word history eventually settles on for its authors, “primitive” is not available.

³¹⁰García Sanjuán, L., et al. — “In the bosom of the Earth: a new megalithic monument at the Antequera World Heritage Site,” *Antiquity* (2023). Piedras Blancas tomb; La Peña's geology and the 45° alignment's cliff-plane target. <https://www.cambridge.org/core/journals/antiquity/article/in-the-bosom-of-the-earth-a-new-megalithic-monument-at-the-antequera-world-heritage-site/2C15E82312640F12CBC729EFE01CA9EB>

³¹¹Medievalists.net — “Two Medieval Men Found Buried in Prehistoric Site” (2026), reporting Silva et al., *Journal of Archaeological Science: Reports* 69 (2026): the axis-aligned 8th–11th c. burials. <https://www.medievalists.net/2026/03/two-medieval-men-found-buried-in-prehistoric-site/>

³¹²García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

³¹³Lozano Rodríguez, J.A., García Sanjuán, L., et al. — “Early science and colossal stone engineering in Menga, a Neolithic dolmen (Antequera, Spain),” *Science Advances* 10, eadp1295 (2024). The geoarchaeological synthesis: angles, polarity, foundations, construction sequence. <https://www.science.org/doi/10.1126/sciadv.adp1295>

³¹⁴Lozano Rodríguez, J.A., García Sanjuán, L., Álvarez-Valero, A.M., et al. — “The provenance of the stones in the Menga dolmen reveals one of the greatest engineering feats of the Neolithic,” *Scientific Reports* 13, 21184 (2023). Quarry identification, five lithologies, fracture analysis, C-5 mass calculation. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10692229/>

³¹⁵University of Alcalá repository record for the *Science Advances* paper, including the official abstract (“no precedents in Iberia”). <https://ebuah.uah.es/dspace/handle/10017/63692?show=full>

Conclusion 4 — Orientation: [MEASURED], resolved. The 45° axis deliberately targets the northern cliff of La Peña de los Enamorados — the only such orientation in continental Europe — overriding the pan-Iberian solar custom at the moment of maximum monumental ambition ³¹⁶³¹⁷. The mountain took priority over the sun. What the mountain *was* to them is [OPEN].

Conclusion 5 — Light: [MEASURED] effect, [INFERRED] instrument. The chamber produces a specific, reproducible summer-solstice light asymmetry, achieved through measured mathematical shaping of the left wall. Combined with the orientation, the chamber is best read as a dual-constraint optical device: pointing at the mountain, harvesting the sun ³¹⁸³¹⁹. The lunar-dial extension is [OPEN], pending survey-grade testing.

Conclusion 6 — Water: [MEASURED] feature, strongly [INFERRED] primordial function. The 19.4 m well is unique in megalithic Europe, architecturally integral, and taps fresh water on a brackish plain. Its suppression in the 18th century and folk-memory as healing water corroborate a continuous, central role. A monument with its own freshwater source is a resource claim in stone — this investigation's strongest single inference about *why this hill* ³²⁰³²¹³²².

Conclusion 7 — Discontinuity: [MEASURED], unexplained. The scale and sophistication appear without local precedent or visible learning curve (Flag 7), at the moment of the region's first great dolmen (Flag 10). The explanations on the table — lost prototypes, Atlantic inspiration, post-El Toro reinvention, crossroads knowledge-hybridization — are individually insufficient and collectively plausible. The discontinuity stands as the book's main open question.

Conclusion 8 — Continuity: [MEASURED/DOCUMENTED]. For a thousand years, successors built on Menga's sight-line; for five thousand, every culture on the plain used, buried at, feared, or healed at the monument. The axis-aligned medieval burials prove the line itself was transmitted across four millennia ³²³³²⁴. Menga is among the longest continuously meaningful structures on Earth.

³¹⁶García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

³¹⁷UNESCO World Heritage Centre — “Antequera Dolmens Site,” inscription 1501 (2016), criteria (iii) and (iv). <https://whc.unesco.org/en/list/1501/>

³¹⁸NovoScriptorium — “The great Menga Dolmen of Antequera”: the left-wall third-degree polynomial fit, left/right asymmetry, 2005 laser-scan program. <https://novoscriptorium.com/2020/03/22/the-great-menga-dolmen-of-antequera-malaga-spain/>

³¹⁹García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

³²⁰García Sanjuán, L., Balsera Nieto, V., et al. — “A multimethod approach to the genesis of Menga, a World Heritage megalith,” *Quaternary Research* (2023). The dating backbone: 29 new numerical ages, Bayesian models, OSL from Pillar 3's socket. <https://www.cambridge.org/core/journals/quaternary-research/article/multimethod-approach-to-the-genesis-of-menga-a-world-heritage-megalith/A6CFB1AE2B583CD57BC1F044F12D66B5>

³²¹García Sanjuán, L., et al. — “Waterscapes through time. The Menga well as a unique hydraulic resource in its geographic and historical context” (2021). The well's excavation, infill chronology, water chemistry, and the brackish-water landscape. <https://www.academia.edu/69337033/>

³²²Hobthrush — “Dolmen de Menga”: the well's early-1700s suppression, ~50 t of fill, animal carcasses, carved crosses, and the pre-suppression clearing hypothesis (tagged [REPORTED]/[INFERRED] in the main text). <https://hobthrush.co.uk/dolmen-de-menga-antequera-spain/>

³²³Medievalists.net — “Two Medieval Men Found Buried in Prehistoric Site” (2026), reporting Silva et al., *Journal of Archaeological Science: Reports* 69 (2026): the axis-aligned 8th–11th c. burials. <https://www.medievalists.net/2026/03/two-medieval-men-found-buried-in-prehistoric-site/>

³²⁴Antequera Tourism — official monument dossier: the two-line orientation system (Menga→La Peña; Viera→equinox; El Romeral on the axis, → El Torcal). <https://turismo.antequera.es/wp-content/uploads/2011/06/DOLMENES-INGLES.pdf>

CHAPTER 25: WHAT REMAINS OPEN — THE HONEST FRONTIER

A professional investigation ends by stating what it could not close. In descending order of consequence:

1. **The referent of the mountain.** La Peña was sacred before Menga (Matacabras predates it ³²⁵) and was worth breaking the solar rule for. Ancestor, sleeper, deity, wayfinding totem — no instrument can decide. [OPEN]
2. **The missing ladder.** Where was the engineering learned? Perishable-media apprenticeship at a crossroads is this investigation's best reading, but a single excavated "practice" monument would transform the question. [OPEN — and findable]
3. **The well's construction date.** Architectural integration argues original; the absence of prehistoric fill leaves formal room for doubt. [OPEN — resolvable in principle by sediment coring at the shaft's rim or micro-stratigraphy of its wall surfaces]
4. **The lunar instrument.** Plan features (R4/R7 steps) are suggestive; no peer-reviewed survey with horizon altitudes has tested the standstill readings. [OPEN — resolvable by a single field campaign]
5. **The primary burial assemblage.** The Neolithic dead of Menga are largely lost to antiquarian clearance; the "hundreds of skeletons" remain [REPORTED]. [OPEN — partially resolvable by archival and museum collection audits]
6. **The fourth pillar and the missing capstone C-0/entrance arrangement.** Their absence changes the light aperture and thus all optical reconstructions. [OPEN]
7. **The fifth lithology's quarry.** One of the five stone types is unmatched to its outcrop. [OPEN — resolvable by continued survey]
8. **The meaning of the inward-facing fossils.** Protective rotation, marine symbolism, or both. [OPEN — perhaps permanently]

CHAPTER 26: EPILOGUE — FOLLOWING THE CLUES TO WHERE THEY STOP

Every investigation has a last page, and this one's is written in the chamber itself, at the hour the builders chose.

Stand inside Menga on a June morning near the solstice. The corridor behind you points 45 degrees northeast, out the door, across the plain, to the chin of the sleeping mountain — the alignment that cost the builders the easy solar geometry every one of their neighbors used, and that they paid for with a reshaped wall and a harder kind of light. At your feet, the floor drops away: nineteen meters of hand-cut shaft into the old seabed, down to the plain's hidden fresh water — the well that fed a village, healed a legend, and frightened an empire's church into burying it under fifty tons of earth and dead animals, and that still holds water today. Around you, the walls lean in, thirty-one great stones and one that buckled, every one of them turned so the memory of the sea faces the people inside. Overhead, a hundred and fifty tons of stone the color of dry bone rests on a curve no one can see, an arch invented by people who had never seen an arch, holding — so far — for fifty-seven centuries.

This investigation began with a list of anomalies and a promise: no adopted hypotheses, follow the clues. The clues led to a place more interesting than any of the ready-made stories. Not a lost civilization — the builders are found; their villages, their caves, their cairns, their pottery are all over the plain. Not primitive simplicity — the measurements forbid it. What the clues actually describe is a moment

³²⁵Wikipedia — "Dolmen of Menga" (reference summary; the "several hundred skeletons" tradition is [REPORTED] through this and the following). https://en.wikipedia.org/wiki/Dolmen_of_Menga

when a community of farmers, at a crossroads, beside a mountain that looked like a sleeping person, over the plain’s only reliable fresh water, in the generations after their ancestral cave’s roof fell in, made a collective decision to build something that would outlast everything they were — and then executed it with a command of stone, water, light, and force that their descendants would not match for three thousand years.

The anomalies of Menga are not evidence of interference. They are evidence of *occasion*: of what happens when necessity, knowledge, grief, water, and a mountain all converge on one hill, and a society finds out — perhaps to its own astonishment — what it can do.

The flags remain in the register. The open questions remain open, and they are stated precisely so that the next instrument aimed at this hill — and there will be one — knows exactly where to point.

The investigation rests. The monument does not.

— *End of the main text. Appendices follow: the complete Flag Register, monument data tables, the timeline, the glossary for general readers, and the full source apparatus with pedigrees.*

APPENDICES

APPENDIX A: THE COMPLETE FLAG AND ANOMALY REGISTER

Every flag documented in this investigation, consolidated. Each entry: the observed anomaly, its evidence tier, where it is investigated in this book, and its current status. Sub-flags carried forward from the preliminary investigation file are preserved and cross-mapped.

#	Flag	Core measured fact	Tier	Chapter	Status
1	The 45° orientation	Axis at azimuth ~45°, ~10–15° north of the solstitial sunrise extreme; 97% of Iberian tombs obey the solar arc	[MEASURED]	11	Resolved as deliberate mountain alignment; referent’ s meaning OPEN
2	The solstice light machine	Summer-solstice light strikes right wall while left stays in shadow; left wall fits a third-degree polynomial; right wall irregular	[MEASURED]	12	Resolved as engineered effect; instrument-reading INFERRED; lunar extension OPEN

Table 2 – continued

#	Flag	Core measured fact	Tier	Chapter	Status
2a	Stepped orthostats R4/R7	Two right-wall stones step forward at positions matching solar/lunar light boundaries in plan-based models	[MEASURED] feature; [REPORTED] interpretation	12.3	OPEN
3	The well	19.4 m × 1.5 m shaft, axially centered behind Pillar 3, reaching fresh aquifer; unique in megalithic Europe	[MEASURED]	13	Original-construction strongly INFERRED; formally OPEN
3a	The well's suppression	Early-1700s backfilling with ~50 t of material incl. whole animal carcasses; carved crosses at entrance	[MEASURED] fill; [REPORTED] motive	13.4	Documented event; motive INFERRED
3b	The fresh-water monopoly	Regional waters predominantly brackish (gypsum/halite geology); the well yields potable water	[MEASURED]	13.2	Resolved as resource significance
4	Stratigraphic polarity	All structural stones rotated so the original seabed “floor” face (fossils, burrows, imbrications) faces the chamber interior; zero documented exceptions	[MEASURED]	14	Rotation deliberate (proves inside-out build); full motive INFERRED

Table 2 – continued

#	Flag	Core measured fact	Tier	Chapter	Status
5	Soft stone on purpose	Entire monument in porous “soft stone” (porosity 13–30%) despite nearby hard rocks used for tools	[MEASURED]	15	Resolved as informed selection + compensating protection system
6	Deep foundations	Sockets 2.75–3.2 m (~ of each stone) in bedrock; “box” design; stepped terrace foundation ditches	[MEASURED]	16	Performance MEASURED (5,700 yr seismic survival); intent INFERRED
6a	O-14 buckling	One orthostat failing at hidden bedding discontinuity under C-5 + tumulus load; structure unaffected	[MEASURED]	16.2	Resolved: single material-selection error absorbed by redundancy
7	The missing ladder	Oldest great Iberian dolmen shows engineering with no known local precedents; no practice-scale monuments identified	[MEASURED]/[DOCUMENTED] absence	17	OPEN —the book’ s main unresolved question
7a	The El Toro adjacency	Regional ancestral cave destroyed by roof collapse c. 4250–3935 BCE, shortly before the project	[MEASURED]	2.2	Chronology settled; causal link OPEN

Table 2 – continued

#	Flag	Core measured fact	Tier	Chapter	Status
8	The thousand-year line	Menga, Viera, El Romeral and La Peña on one visual axis; four calendar events keyed across ~1,000 years	[MEASURED]/[DOCUMENTED]	18	Resolved as maintained inter-monument design; transmission mechanism OPEN
9	Tomb vs. machine	Engineering features (well, light, axis, seismic redundancy) exceed burial function; primary burial assemblage poorly preserved	[MEASURED] features; [REPORTED] skeletons	19	Resolved as multi-function monument; burial scale OPEN
9a	“Several hundred skeletons”	19th-century accounts of mass human remains at first opening	[REPORTED]	19.1	Unverifiable today; must be quoted with pedigree tag
10	Scale extremes	C-5 ~150 t (Iberian record, 2nd in Europe); total ~1,140 t; largest European stone monument of its time —and the <i>first</i> of its class	[MEASURED]	20	Resolved; anomalous in primacy, corroborates Flag 7
11	The buried-build inversion	Chamber built in a trench; capstones slid at ground level; floor lowered afterward —no lifting of the 150-t roof	[MEASURED] geometry → [INFERRED] sequence	7, 9.4	Resolved as construction method

Table 2 – continued					
#	Flag	Core measured fact	Tier	Chapter	Status
12	Millimetric interlock	Facet angles 77–80°; lean held at 85.2°±1.6°/84.0° ±2.0° across 20+ stones; custom wedge-profiles per stone	[MEASURED]	8	Resolved; implies standard gauges/templates
13	The armored tumulus	Alternating locked sandstone slabs and rammed clay over 3 m; waterproof; expanded after initial construction	[MEASURED]	10	Resolved as protection system
14	Axis-aligned medieval burials	Two 8th–11th c. inhumations in the atrium, 190 years apart, both aligned to the monument’ s axis	[MEASURED]	21.2	Resolved as continued axis knowledge 4,000+ years on
15	Absence flags	No metal tools, wheels, or draft animals attested for the construction phase; the entire logistics package is wood-fiber-stone	[MEASURED] absence	5–6	Resolved as constraint on all reconstructions

APPENDIX B: THE MONUMENT IN NUMBERS

Consolidated data table. All values [MEASURED] unless noted.

Setting

Parameter	Value
Location	Antequera, Málaga province, Andalusia, Spain; ~37°N
Plain elevation	~400–450 m asl
La Peña de los Enamorados	878–880 m asl; chin cliff ~100 m high
El Torcal	11 km south; 1,100–1,400 m karst plateau
Quarry (Cerro de la Cruz)	~0.85–1 km distant; ~50–100 m above the site
Transport slope	~22° average, continuously downhill

The chamber

Parameter	Value
Gallery length	24.9 m (interior); ~27.5 m with atrium approaches
Width	max ~5.7–6 m (rear third)
Interior height	2.5 m (entrance) → 3.45 m (back)
Orthostats	10 per side in the chamber + backstone (O-12) + entrance orthostats
Capstones	5 surviving; a 6th (entrance) missing
Pillars	3 surviving on the axis; a 4th probable, missing
Total stones	32; total mass ~1,140 t
Largest stone	C-5: $6.95 \times 6 \times 1.88$ m; $\sim 149.6 \pm 9.2$ t
Wall lean	left $85.2^\circ \pm 1.6^\circ$; right $84.0^\circ \pm 2.0^\circ$ from horizontal
Facet interlock angles	~77–80°
Foundation embedment	2.75 m (entrance stones) → 3.20 m (rear stones); ~ of stone length
Axis azimuth	~45° (NE), to La Peña' s northern cliff
Well	19.4 m deep; ~1.5 m mouth diameter; reaches fresh aquifer

The mound

Parameter	Value
Dimensions	48.3 m (NE–SW) $\times$ 44 m (NW–SE); ~2,730 m ²
Structure	basal activity layer ~1.5 m; engineered body ~3 m of alternating slabs and rammed clay

Dating

Method	Result
Radiocarbon (mound stratigraphy, 11–12 determinations, Bayesian)	construction window 3835–3715 to 3575–3500 BCE (1 , model-dependent); span 140–275 yr
OSL (Pillar 3 socket infill, 3 ages)	3769 ± 201 / 3517 ± 325 / 3543 ± 510 BCE
Consensus construction window	c. 3800–3600 BCE
El Toro cave abandonment	c. 4250–3935 BCE (2)

Table 6 – continued

Method	Result
Arroyo Saladillo cairn burial	4037–3805 BCE (2)
Viera ante quem date	GrN-16067: 4550 ± 140 BP
Well infill	15th–18th c. CE (16 determinations)
Medieval atrium burials	8th–11th c. CE, ~190 yr apart

APPENDIX C: TIMELINE

When	Event
150–200 million years ago	El Torcal’ s Jurassic limestone laid down in the Tethys Ocean
~8–11 million years ago (Tortonian)	The future Menga stones form as seabed and submarine-canyon sediments
c. 5400 BCE	First Neolithic farmers occupy El Toro cave
c. 4700 BCE	Great megalithic mounds rise in Brittany —the Atlantic precedent
c. 4300–3800 BCE	Intensified Late Neolithic occupation across the Antequera district
c. 4250–3935 BCE	El Toro cave destroyed by roof collapse; abandonment
4037–3805 BCE	Proto-megalithic cairn burial at Arroyo Saladillo
before c. 3800 BCE	Matacabras rock-art sanctuary in use on La Peña’ s northern face
c. 3800–3600 BCE	Construction of Menga
after Menga	Viera built 70 m away, facing the equinox sunrise
c. 3rd millennium BCE	Piedras Blancas tomb in use in the giant’ s chest; El Romeral tholos built on the Menga–La Peña sight-line, facing El Torcal; winter-solstice light keyed
c. 1950–1180 BCE	Piedras Blancas abandoned after three use phases
Bronze & Iron Ages	Continued use-deposits at Menga
Roman period	Tombs and structures around the dolmen
8th–11th c. CE	Two axis-aligned male burials in the atrium
1410	Christian conquest of Antequera; the monument’ s medieval legend era begins
1492	Columbus compares Cuban mountains to “la peña de las enamoradas”
1530	First written record: <i>antrum de Menga</i> in a Málaga bishop’ s document
late 1500s	Reports of “pagan” practices at the Cueva de Menga

Table 7 – continued

When	Event
early 1700s	The well backfilled with ~50 t of material incl. animal carcasses; crosses carved at the entrance; site suppressed
1842–1847	Mitjana excavates; 1847 <i>Memoria</i> (“druid temple”); backstone perforation
1851–1853	Louisa Tenison records the Menga–La Peña alignment and the open well
c. 1850–1868	Well re-backfilled and forgotten for 137 years
1874	Trinidad de Rojas publishes the Menga legend in <i>Genil</i>
1903–1905	Viera brothers excavate Viera and locate El Romeral
1934	Hemp publishes the first photograph of La Peña from Menga’ s entrance
1940s	Vehicle track cuts into the mound
1980s–1991	University of Málaga excavations; C-5’ s full convex form photographed
2002	Hoskin instrumentally confirms the three orientations
2005	Full 3D laser scan of Menga; well shaft rediscovered and excavated to 19.4 m
2005–2006	University of Granada excavations (mound, atrium, well)
15 July 2016	UNESCO World Heritage inscription: three monuments + two natural monuments
2018	Matacabras rock art dated and analyzed (ATLAS group)
2020	Piedras Blancas tomb excavated
2023	Multimethod dating paper (3800–3600 BCE); quarry provenance paper (Cerro de la Cruz)
2024	<i>Science Advances</i> geoarchaeological synthesis: early science, colossal engineering
2026	Archaeogenetic study of the medieval burials published

APPENDIX D: GLOSSARY FOR GENERAL READERS

- **Orthostat:** a large upright stone slab forming the wall of a megalithic chamber.
- **Capstone:** a massive horizontal slab forming the roof. Menga’s are numbered C-1 (entrance) to C-5 (rear).
- **Backstone:** the single slab closing the rear of the chamber.
- **Dolmen:** a megalithic tomb of uprights roofed by capstones, originally covered by a mound.

- **Passage grave:** a dolmen with a distinct entrance corridor (Viera). Menga, a “simple gallery,” barely separates corridor from chamber.
- **Tholos:** a round chamber roofed by a corbelled false dome of inward-stepping stone rings (El Romeral).
- **Tumulus:** the earthen mound covering a tomb. Menga’s is an engineered composite, not a soil heap.
- **Calcarenite / calcirudite / breccia:** sedimentary rocks made of carbonate sand, gravel-sized debris, and angular fragments respectively — all “soft stones” in engineering terms.
- **Porosity:** the percentage of empty space in rock; high porosity means the rock absorbs water.
- **Tectonic joints:** natural fracture planes in bedrock, exploited by the builders as ready-made block boundaries.
- **Stratigraphic polarity:** the “which way was up” of a sedimentary rock, readable from fossils, burrows, and pebble arrangements.
- **Relieving arch:** a structural form that redirects load outward to supports, sparing the span’s center. Menga’s convex capstone is the earliest known example.
- **Plumb bob:** a weight on a cord giving true vertical — the probable master instrument of the build.
- **Azimuth:** compass bearing in degrees (0° = north, 90° = east). Sunrise at Antequera ranges roughly 60° (June) to 120° (December).
- **Summer solstice:** the year’s longest day (~21 June); the sun rises at its northernmost point.
- **Lunar standstill:** the moon’s 18.6-year cycle of extreme northern/southern rising positions.
- **Radiocarbon dating:** age measurement from the decay of carbon-14 in once-living material.
- **OSL (optically stimulated luminescence) dating:** measures when buried mineral grains last saw sunlight — used on the sediments inside Pillar 3’s socket.
- **Bayesian modeling:** statistical combination of many dates with stratigraphic order, producing tighter ranges than any single date.
- **Terminus post quem / ante quem:** “date after which” / “date before which” brackets provided by dated layers below/above a structure.
- **Geoarchaeology:** the combined reading of a site through geology and archaeology together — the method behind the 2024 synthesis.
- **[MEASURED] / [DOCUMENTED] / [INFERRED] / [REPORTED] / [OPEN]:** this investigation’s evidence pedigrees — see the Foreword.

APPENDIX E: SOURCES AND PEDIGREES

Every numbered citation in this book. Primary scientific literature listed first; institutional and historical sources next; popular and traditional accounts last, each with its reliability note. Where a claim in the main text rests on a popular source alone, it is tagged [REPORTED] in place.

Primary scientific literature

Institutional and heritage sources

Science journalism (used for reportage of the primary literature, cross-checked against it)

Site histories, geometry studies, and reference works

Folklore and legend studies ([REPORTED] tier throughout)

Independent archaeoastronomy (non-peer-reviewed; treated as [OPEN] hypothesis material only)

Investigation compiled August 2026. All [MEASURED] values quoted from the cited primary literature; where sources conflict (e.g., capstone weight estimates of 150 vs. 180 tons), both figures and their provenances are given in the text. Nothing in this book has been omitted for convenience, and no hypothesis has been adopted as the investigator's own except where explicitly labeled as this investigation's inference from the measured record.